

For the following exercises, find the requested higher-order derivative for the given functions.

209. $\frac{d^3 y}{dx^3}$ of $y = 3 \cos x$

210. $\frac{d^2 y}{dx^2}$ of
 $y = 3 \sin x + x^2 \cos x$

211. $\frac{d^4 y}{dx^4}$ of $y = 5 \cos x$

212. $\frac{d^2 y}{dx^2}$ of $y = \sec x + \cot x$

213. $\frac{d^3 y}{dx^3}$ of $y = x^{10} - \sec x$

3.6 The Chain Rule

Learning Objectives

- 3.6.1 State the chain rule for the composition of two functions.
- 3.6.2 Apply the chain rule together with the power rule.
- 3.6.3 Apply the chain rule and the product/quotient rules correctly in combination when both are necessary.
- 3.6.4 Recognize the chain rule for a composition of three or more functions.
- 3.6.5 Describe the proof of the chain rule.

We have seen the techniques for differentiating basic functions (x^n , $\sin x$, $\cos x$, etc.) as well as sums, differences, products, quotients, and constant multiples of these functions. However, these techniques do not allow us to differentiate compositions of functions, such as $h(x) = \sin(x^3)$ or $k(x) = \sqrt{3x^2 + 1}$. In this section, we study the rule for finding the derivative of the composition of two or more functions.

Deriving the Chain Rule

When we have a function that is a composition of two or more functions, we could use all of the techniques we have already learned to differentiate it. However, using all of those techniques to break down a function into simpler parts that we are able to differentiate can get cumbersome. Instead, we use the **chain rule**, which states that the derivative of a composite function is the derivative of the outer function evaluated at the inner function times the derivative of the inner function.

To put this rule into context, let's take a look at an example: $h(x) = \sin(x^3)$. We can think of the derivative of this function with respect to x as the rate of change of $\sin(x^3)$ relative to the change in x . Consequently, we want to know how $\sin(x^3)$ changes as x changes. We can think of this event as a chain reaction: As x changes, x^3 changes, which leads to a change in $\sin(x^3)$. This chain reaction gives us hints as to what is involved in computing the derivative of $\sin(x^3)$. First of all, a change in x forcing a change in x^3 suggests that somehow the derivative of x^3 is involved. In addition, the change in x^3 forcing a change in $\sin(x^3)$ suggests that the derivative of $\sin(u)$ with respect to u , where $u = x^3$, is also part of the final derivative.

We can take a more formal look at the derivative of $h(x) = \sin(x^3)$ by setting up the limit that would give us the derivative at a specific value a in the domain of $h(x) = \sin(x^3)$.

$$h'(a) = \lim_{x \rightarrow a} \frac{\sin(x^3) - \sin(a^3)}{x - a}.$$

This expression does not seem particularly helpful; however, we can modify it by multiplying and dividing by the expression $x^3 - a^3$ to obtain

$$h'(a) = \lim_{x \rightarrow a} \frac{\sin(x^3) - \sin(a^3)}{x^3 - a^3} \cdot \frac{x^3 - a^3}{x - a}.$$

From the definition of the derivative, we can see that the second factor is the derivative of x^3 at $x = a$. That is,

$$\lim_{x \rightarrow a} \frac{x^3 - a^3}{x - a} = \frac{d}{dx}(x^3)_{x=a} = 3a^2.$$

However, it might be a little more challenging to recognize that the first term is also a derivative. We can see this by letting $u = x^3$ and observing that as $x \rightarrow a$, $u \rightarrow a^3$:

$$\begin{aligned} \lim_{x \rightarrow a} \frac{\sin(x^3) - \sin(a^3)}{x^3 - a^3} &= \lim_{u \rightarrow a^3} \frac{\sin u - \sin(a^3)}{u - a^3} \\ &= \frac{d}{du}(\sin u)_{u=a^3} \\ &= \cos(a^3). \end{aligned}$$

Thus, $h'(a) = \cos(a^3) \cdot 3a^2$.

In other words, if $h(x) = \sin(x^3)$, then $h'(x) = \cos(x^3) \cdot 3x^2$. Thus, if we think of $h(x) = \sin(x^3)$ as the composition $(f \circ g)(x) = f(g(x))$ where $f(x) = \sin x$ and $g(x) = x^3$, then the derivative of $h(x) = \sin(x^3)$ is the product of the derivative of $g(x) = x^3$ and the derivative of the function $f(x) = \sin x$ evaluated at the function $g(x) = x^3$. At this point, we anticipate that for $h(x) = \sin(g(x))$, it is quite likely that $h'(x) = \cos(g(x))g'(x)$. As we determined above, this is the case for $h(x) = \sin(x^3)$.

Now that we have derived a special case of the chain rule, we state the general case and then apply it in a general form to other composite functions. An informal proof is provided at the end of the section.

Rule: The Chain Rule

Let f and g be functions. For all x in the domain of g for which g is differentiable at x and f is differentiable at $g(x)$, the derivative of the composite function

$$h(x) = (f \circ g)(x) = f(g(x))$$

is given by

$$h'(x) = f'(g(x))g'(x). \quad (3.17)$$

Alternatively, if y is a function of u , and u is a function of x , then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

MEDIA

Watch an [animation \(http://www.openstax.org/l/20_chainrule2\)](http://www.openstax.org/l/20_chainrule2) of the chain rule.

Problem-Solving Strategy

Applying the Chain Rule

1. To differentiate $h(x) = f(g(x))$, begin by identifying $f(x)$ and $g(x)$.
2. Find $f'(x)$ and evaluate it at $g(x)$ to obtain $f'(g(x))$.
3. Find $g'(x)$.
4. Write $h'(x) = f'(g(x)) \cdot g'(x)$.

Note: When applying the chain rule to the composition of two or more functions, keep in mind that we work our way from the outside function in. It is also useful to remember that the derivative of the composition of two functions can be thought of as having two parts; the derivative of the composition of three functions has three parts; and so on. Also, remember that we never evaluate a derivative at a derivative.

The Chain and Power Rules Combined

We can now apply the chain rule to composite functions, but note that we often need to use it with other rules. For example, to find derivatives of functions of the form $h(x) = (g(x))^n$, we need to use the chain rule combined with the power rule. To do so, we can think of $h(x) = (g(x))^n$ as $f(g(x))$ where $f(x) = x^n$. Then $f'(x) = nx^{n-1}$. Thus, $f'(g(x)) = n(g(x))^{n-1}$. This leads us to the derivative of a power function using the chain rule,

$$h'(x) = n(g(x))^{n-1}g'(x)$$

Rule: Power Rule for Composition of Functions

For all values of x for which the derivative is defined, if

$$h(x) = (g(x))^n.$$

Then

$$h'(x) = n(g(x))^{n-1} g'(x). \quad (3.18)$$

EXAMPLE 3.48

Using the Chain and Power Rules

Find the derivative of $h(x) = \frac{1}{(3x^2+1)^2}$.

Solution

First, rewrite $h(x) = \frac{1}{(3x^2+1)^2} = (3x^2 + 1)^{-2}$.

Applying the power rule with $g(x) = 3x^2 + 1$, we have

$$h'(x) = -2(3x^2 + 1)^{-3} (6x).$$

Rewriting back to the original form gives us

$$h'(x) = \frac{-12x}{(3x^2 + 1)^3}.$$

 3.34 Find the derivative of $h(x) = (2x^3 + 2x - 1)^4$.

EXAMPLE 3.49

Using the Chain and Power Rules with a Trigonometric Function

Find the derivative of $h(x) = \sin^3 x$.

Solution

First recall that $\sin^3 x = (\sin x)^3$, so we can rewrite $h(x) = \sin^3 x$ as $h(x) = (\sin x)^3$.

Applying the power rule with $g(x) = \sin x$, we obtain

$$h'(x) = 3(\sin x)^2 \cos x = 3 \sin^2 x \cos x.$$

EXAMPLE 3.50

Finding the Equation of a Tangent Line

Find the equation of a line tangent to the graph of $h(x) = \frac{1}{(3x-5)^2}$ at $x = 2$.

Solution

Because we are finding an equation of a line, we need a point. The x -coordinate of the point is 2. To find the y -coordinate, substitute 2 into $h(x)$. Since $h(2) = \frac{1}{(3(2)-5)^2} = 1$, the point is $(2, 1)$.

For the slope, we need $h'(2)$. To find $h'(x)$, first we rewrite $h(x) = (3x - 5)^{-2}$ and apply the power rule to obtain

$$h'(x) = -2(3x - 5)^{-3} (3) = -6(3x - 5)^{-3}.$$

By substituting, we have $h'(2) = -6(3(2) - 5)^{-3} = -6$. Therefore, the line has equation $y - 1 = -6(x - 2)$. Rewriting, the equation of the line is $y = -6x + 13$.

 3.35 Find an equation of the line tangent to the graph of $f(x) = (x^2 - 2)^3$ at $x = -2$.

Combining the Chain Rule with Other Rules

Now that we can combine the chain rule and the power rule, we examine how to combine the chain rule with the other rules we have learned. In particular, we can use it with the formulas for the derivatives of trigonometric functions or with the product rule.

EXAMPLE 3.51

Using the Chain Rule on a General Cosine Function

Find the derivative of $h(x) = \cos(g(x))$.

✓ Solution

Think of $h(x) = \cos(g(x))$ as $f(g(x))$ where $f(x) = \cos x$. Since $f'(x) = -\sin x$, we have $f'(g(x)) = -\sin(g(x))$. Then we do the following calculation.

$$\begin{aligned} h'(x) &= f'(g(x)) g'(x) && \text{Apply the chain rule.} \\ &= -\sin(g(x)) g'(x) && \text{Substitute } f'(g(x)) = -\sin(g(x)). \end{aligned}$$

Thus, the derivative of $h(x) = \cos(g(x))$ is given by $h'(x) = -\sin(g(x)) g'(x)$.

In the following example we apply the rule that we have just derived.

EXAMPLE 3.52

Using the Chain Rule on a Cosine Function

Find the derivative of $h(x) = \cos(5x^2)$.

✓ Solution

Let $g(x) = 5x^2$. Then $g'(x) = 10x$. Using the result from the previous example,

$$\begin{aligned} h'(x) &= -\sin(5x^2) \cdot 10x \\ &= -10x \sin(5x^2). \end{aligned}$$

EXAMPLE 3.53

Using the Chain Rule on Another Trigonometric Function

Find the derivative of $h(x) = \sec(4x^5 + 2x)$.

✓ Solution

Apply the chain rule to $h(x) = \sec(g(x))$ to obtain

$$h'(x) = \sec(g(x)) \tan(g(x)) g'(x).$$

In this problem, $g(x) = 4x^5 + 2x$, so we have $g'(x) = 20x^4 + 2$. Therefore, we obtain

$$\begin{aligned} h'(x) &= \sec(4x^5 + 2x) \tan(4x^5 + 2x) (20x^4 + 2) \\ &= (20x^4 + 2) \sec(4x^5 + 2x) \tan(4x^5 + 2x). \end{aligned}$$

✓ 3.36 Find the derivative of $h(x) = \sin(7x + 2)$.

At this point we provide a list of derivative formulas that may be obtained by applying the chain rule in conjunction with the formulas for derivatives of trigonometric functions. Their derivations are similar to those used in [Example 3.51](#) and [Example 3.53](#). For convenience, formulas are also given in Leibniz's notation, which some students find easier to remember. (We discuss the chain rule using Leibniz's notation at the end of this section.) It is not absolutely necessary to memorize these as separate formulas as they are all applications of the chain rule to previously learned formulas.

Theorem 3.10**Using the Chain Rule with Trigonometric Functions**

For all values of x for which the derivative is defined,

$$\begin{array}{ll}
 \frac{d}{dx}(\sin(g(x))) = \cos(g(x)) g'(x) & \frac{d}{dx} \sin u = \cos u \frac{du}{dx} \\
 \frac{d}{dx}(\cos(g(x))) = -\sin(g(x)) g'(x) & \frac{d}{dx} \cos u = -\sin u \frac{du}{dx} \\
 \frac{d}{dx}(\tan(g(x))) = \sec^2(g(x)) g'(x) & \frac{d}{dx} \tan u = \sec^2 u \frac{du}{dx} \\
 \frac{d}{dx}(\cot(g(x))) = -\csc^2(g(x)) g'(x) & \frac{d}{dx} \cot u = -\csc^2 u \frac{du}{dx} \\
 \frac{d}{dx}(\sec(g(x))) = \sec(g(x)) \tan(g(x)) g'(x) & \frac{d}{dx} \sec u = \sec u \tan u \frac{du}{dx} \\
 \frac{d}{dx}(\csc(g(x))) = -\csc(g(x)) \cot(g(x)) g'(x) & \frac{d}{dx} \csc u = -\csc u \cot u \frac{du}{dx}
 \end{array}$$

EXAMPLE 3.54**Combining the Chain Rule with the Product Rule**

Find the derivative of $h(x) = (2x + 1)^5(3x - 2)^7$.

✓ Solution

First apply the product rule, then apply the chain rule to each term of the product.

$$\begin{aligned}
 h'(x) &= \frac{d}{dx}((2x + 1)^5) \cdot (3x - 2)^7 + \frac{d}{dx}((3x - 2)^7) \cdot (2x + 1)^5 && \text{Apply the product rule.} \\
 &= 5(2x + 1)^4 \cdot 2 \cdot (3x - 2)^7 + 7(3x - 2)^6 \cdot 3 \cdot (2x + 1)^5 && \text{Apply the chain rule.} \\
 &= 10(2x + 1)^4(3x - 2)^7 + 21(3x - 2)^6(2x + 1)^5 && \text{Simplify.} \\
 &= (2x + 1)^4(3x - 2)^6(10(3x - 2) + 21(2x + 1)) && \text{Factor out } (2x + 1)^4(3x - 2)^6. \\
 &= (2x + 1)^4(3x - 2)^6(72x + 1) && \text{Simplify.}
 \end{aligned}$$

✓ 3.37 Find the derivative of $h(x) = \frac{x}{(2x+3)^3}$.

Composites of Three or More Functions

We can now combine the chain rule with other rules for differentiating functions, but when we are differentiating the composition of three or more functions, we need to apply the chain rule more than once. If we look at this situation in general terms, we can generate a formula, but we do not need to remember it, as we can simply apply the chain rule multiple times.

In general terms, first we let

$$k(x) = h(f(g(x))).$$

Then, applying the chain rule once we obtain

$$k'(x) = \frac{d}{dx}(h(f(g(x)))) = h'(f(g(x))) \cdot \frac{d}{dx}f(g(x)).$$

Applying the chain rule again, we obtain

$$k'(x) = h'(f(g(x))) f'(g(x)) g'(x).$$

Rule: Chain Rule for a Composition of Three Functions

For all values of x for which the function is differentiable, if

$$k(x) = h(f(g(x))),$$

then

$$k'(x) = h'(f(g(x))) f'(g(x)) g'(x).$$

In other words, we are applying the chain rule twice.

Notice that the derivative of the composition of three functions has three parts. (Similarly, the derivative of the composition of four functions has four parts, and so on.) Also, *remember, we can always work from the outside in, taking one derivative at a time.*

EXAMPLE 3.55

Differentiating a Composite of Three Functions

Find the derivative of $k(x) = \cos^4(7x^2 + 1)$.

✓ Solution

First, rewrite $k(x)$ as

$$k(x) = (\cos(7x^2 + 1))^4.$$

Then apply the power rule several times.

$$\begin{aligned} k'(x) &= 4(\cos(7x^2 + 1))^3 \left(\frac{d}{dx} \cos(7x^2 + 1) \right) && \text{Apply the chain rule.} \\ &= 4(\cos(7x^2 + 1))^3 (-\sin(7x^2 + 1)) \left(\frac{d}{dx} (7x^2 + 1) \right) && \text{Apply the chain rule.} \\ &= 4(\cos(7x^2 + 1))^3 (-\sin(7x^2 + 1)) (14x) && \text{Apply the chain rule.} \\ &= -56x \sin(7x^2 + 1) \cos^3(7x^2 + 1) && \text{Simplify.} \end{aligned}$$

✓ 3.38 Find the derivative of $h(x) = \sin^6(x^3)$.

EXAMPLE 3.56

Using the Chain Rule in a Velocity Problem

A particle moves along a coordinate axis. Its position at time t is given by $s(t) = \sin(2t) + \cos(3t)$. What is the velocity of the particle at time $t = \frac{\pi}{6}$?

✓ Solution

To find $v(t)$, the velocity of the particle at time t , we must differentiate $s(t)$. Thus,

$$v(t) = s'(t) = 2\cos(2t) - 3\sin(3t).$$

Substituting $t = \frac{\pi}{6}$ into $v(t)$, we obtain $v\left(\frac{\pi}{6}\right) = -2$.

✓ 3.39 A particle moves along a coordinate axis. Its position at time t is given by $s(t) = \sin(4t)$. Find its acceleration at time t .

Proof

At this point, we present a very informal proof of the chain rule. For simplicity's sake we ignore certain issues: For example, we assume that $g(x) \neq g(a)$ for $x \neq a$ in some open interval containing a . We begin by applying the limit definition of the derivative to the function $h(x)$ to obtain $h'(a)$:

$$h'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a}.$$

Rewriting, we obtain

$$h'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}.$$

Although it is clear that

$$\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} = g'(a),$$

it is not obvious that

$$\lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} = f'(g(a)).$$

To see that this is true, first recall that since g is differentiable at a , g is also continuous at a . Thus,

$$\lim_{x \rightarrow a} g(x) = g(a).$$

Next, make the substitution $y = g(x)$ and $b = g(a)$ and use change of variables in the limit to obtain

$$\lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} = \lim_{y \rightarrow b} \frac{f(y) - f(b)}{y - b} = f'(b) = f'(g(a)).$$

Finally,

$$h'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} = f'(g(a)) g'(a).$$

□

EXAMPLE 3.57

Using the Chain Rule with Functional Values

Let $h(x) = f(g(x))$. If $g(1) = 4$, $g'(1) = 3$, and $f'(4) = 7$, find $h'(1)$.

Solution

Use the chain rule, then substitute.

$h'(1) = f'(g(1)) g'(1)$	Apply the chain rule.
$= f'(4) \cdot 3$	Substitute $g(1) = 4$ and $g'(1) = 3$.
$= 7 \cdot 3$	Substitute $f'(4) = 7$.
$= 21$	Simplify.

 3.40 Given $h(x) = f(g(x))$. If $g(2) = -3$, $g'(2) = 4$, and $f'(-3) = 7$, find $h'(2)$.

The Chain Rule Using Leibniz's Notation

As with other derivatives that we have seen, we can express the chain rule using Leibniz's notation. This notation for the chain rule is used heavily in physics applications.

For $h(x) = f(g(x))$, let $u = g(x)$ and $y = h(x) = f(u)$. Thus,

$$h'(x) = \frac{dy}{dx}, f'(g(x)) = f'(u) = \frac{dy}{du} \text{ and } g'(x) = \frac{du}{dx}.$$

Consequently,

$$\frac{dy}{dx} = h'(x) = f'(g(x)) g'(x) = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Rule: Chain Rule Using Leibniz's Notation

If y is a function of u , and u is a function of x , then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

EXAMPLE 3.58**Taking a Derivative Using Leibniz's Notation, Example 1**

Find the derivative of $y = \left(\frac{x}{3x+2}\right)^5$.

 Solution

First, let $u = \frac{x}{3x+2}$. Thus, $y = u^5$. Next, find $\frac{du}{dx}$ and $\frac{dy}{du}$. Using the quotient rule,

$$\frac{du}{dx} = \frac{2}{(3x+2)^2}$$

and

$$\frac{dy}{du} = 5u^4.$$

Finally, we put it all together.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= 5u^4 \cdot \frac{2}{(3x+2)^2} \\ &= 5\left(\frac{x}{3x+2}\right)^4 \cdot \frac{2}{(3x+2)^2} \\ &= \frac{10x^4}{(3x+2)^6} \end{aligned}$$

Apply the chain rule.

Substitute $\frac{dy}{du} = 5u^4$ and $\frac{du}{dx} = \frac{2}{(3x+2)^2}$.

Substitute $u = \frac{x}{3x+2}$.

Simplify.

It is important to remember that, when using the Leibniz form of the chain rule, the final answer must be expressed entirely in terms of the original variable given in the problem.

EXAMPLE 3.59**Taking a Derivative Using Leibniz's Notation, Example 2**

Find the derivative of $y = \tan(4x^2 - 3x + 1)$.

 Solution

First, let $u = 4x^2 - 3x + 1$. Then $y = \tan u$. Next, find $\frac{du}{dx}$ and $\frac{dy}{du}$:

$$\frac{du}{dx} = 8x - 3 \text{ and } \frac{dy}{du} = \sec^2 u.$$

Finally, we put it all together.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} && \text{Apply the chain rule.} \\ &= \sec^2 u \cdot (8x - 3) && \text{Use } \frac{du}{dx} = 8x - 3 \text{ and } \frac{dy}{du} = \sec^2 u. \\ &= \sec^2(4x^2 - 3x + 1) \cdot (8x - 3) && \text{Substitute } u = 4x^2 - 3x + 1. \end{aligned}$$

-  3.41 Use Leibniz's notation to find the derivative of $y = \cos(x^3)$. Make sure that the final answer is expressed entirely in terms of the variable x .

**SECTION 3.6 EXERCISES**

For the following exercises, given $y = f(u)$ and $u = g(x)$, find $\frac{dy}{dx}$ by using Leibniz's notation for the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

214. $y = 3u - 6, u = 2x^2$

215. $y = 6u^3, u = 7x - 4$

216. $y = \sin u, u = 5x - 1$

217. $y = \cos u, u = \frac{-x}{8}$

218. $y = \tan u, u = 9x + 2$

219. $y = \sqrt{4u + 3}, u = x^2 - 6x$

For each of the following exercises,

- a. decompose each function in the form $y = f(u)$ and $u = g(x)$, and
 b. find $\frac{dy}{dx}$ as a function of x .

220. $y = (3x - 2)^6$

221. $y = (3x^2 + 1)^3$

222. $y = \sin^5(x)$

223. $y = \left(\frac{x}{7} + \frac{7}{x}\right)^7$

224. $y = \tan(\sec x)$

225. $y = \csc(\pi x + 1)$

226. $y = \cot^2 x$

227. $y = -6(\sin x)^{-3}$

For the following exercises, find $\frac{dy}{dx}$ for each function.

228. $y = (3x^2 + 3x - 1)^4$

229. $y = (5 - 2x)^{-2}$

230. $y = \cos^3(\pi x)$

231. $y = (2x^3 - x^2 + 6x + 1)^3$

232. $y = \frac{1}{\sin^2(x)}$

233. $y = (\tan x + \sin x)^{-3}$

234. $y = x^2 \cos^4 x$

235. $y = \sin(\cos 7x)$

236. $y = \sqrt{6 + \sec \pi x^2}$

237. $y = \cot^3(4x + 1)$

238. Let $y = [f(x)]^2$ and suppose that $f'(1) = 4$ and $\frac{dy}{dx} = 10$ for $x = 1$. Find $f(1)$.

239. Let $y = (f(x) + 5x^2)^4$ and suppose that $f(-1) = -4$ and $\frac{dy}{dx} = 3$ when $x = -1$. Find $f'(-1)$.

240. Let $y = (f(u) + 3x)^2$ and $u = x^3 - 2x$. If $f(4) = 6$ and $\frac{dy}{dx} = 18$ when $x = 2$, find $f'(4)$.

241. [T] Find an equation of the tangent line to $y = -\sin\left(\frac{x}{2}\right)$ at the origin. Use a calculator to graph the function and the tangent line together.

242. [T] Find an equation of the tangent line to $y = \left(3x + \frac{1}{x}\right)^2$ at the point $(1, 16)$. Use a calculator to graph the function and the tangent line together.

243. Find the x -coordinates at which the tangent line to $y = \left(x - \frac{6}{x}\right)^8$ is horizontal.

244. [T] Find an equation of the line that is normal to $g(\theta) = \sin^2(\pi\theta)$ at the point $\left(\frac{1}{4}, \frac{1}{2}\right)$. Use a calculator to graph the function and the normal line together.

For the following exercises, use the information in the following table to find $h'(a)$ at the given value for a .

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	2	5	0	2
1	1	-2	3	0
2	4	4	1	-1
3	3	-3	2	3

245. $h(x) = f(g(x)); a = 0$

246. $h(x) = g(f(x)); a = 0$

247. $h(x) = (x^4 + g(x))^{-2}; a = 1$

248. $h(x) = \left(\frac{f(x)}{g(x)} \right)^2$; $a = 3$ 249. $h(x) = f(x + f(x))$; $a = 1$ 250. $h(x) = (1 + g(x))^3$; $a = 2$
251. $h(x) = g(2 + f(x^2))$; $a = 1$ 252. $h(x) = f(g(\sin x))$; $a = 0$ 253. [T] The position function of a freight train is given by $s(t) = 100(t + 1)^{-2}$, with s in meters and t in seconds. At time $t = 6$ s, find the train's
- velocity and
 - acceleration.
 - Using a. and b. is the train speeding up or slowing down?
254. [T] A mass hanging from a vertical spring is in simple harmonic motion as given by the following position function, where t is measured in seconds and s is in inches:
 $s(t) = -3\cos\left(\pi t + \frac{\pi}{4}\right)$.
- Determine the position of the spring at $t = 1.5$ s.
 - Find the velocity of the spring at $t = 1.5$ s.
255. [T] The total cost to produce x boxes of Thin Mint Girl Scout cookies is C dollars, where
 $C = 0.0001x^3 - 0.02x^2 + 3x + 300$.
 In t weeks production is estimated to be $x = 1600 + 100t$ boxes.
- Find the marginal cost $C'(x)$.
 - Use Leibniz's notation for the chain rule, $\frac{dC}{dt} = \frac{dC}{dx} \cdot \frac{dx}{dt}$, to find the rate with respect to time t that the cost is changing.
 - Use b. to determine how fast costs are increasing when $t = 2$ weeks. Include units with the answer.
256. [T] The formula for the area of a circle is $A = \pi r^2$, where r is the radius of the circle. Suppose a circle is expanding, meaning that both the area A and the radius r (in inches) are expanding.
- Suppose $r = 2 - \frac{100}{(t+7)^2}$ where t is time in seconds. Use the chain rule $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$ to find the rate at which the area is expanding.
 - Use a. to find the rate at which the area is expanding at $t = 4$ s.
257. [T] The formula for the volume of a sphere is $S = \frac{4}{3}\pi r^3$, where r (in feet) is the radius of the sphere. Suppose a spherical snowball is melting in the sun.
- Suppose $r = \frac{1}{(t+1)^2} - \frac{1}{12}$ where t is time in minutes. Use the chain rule $\frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dt}$ to find the rate at which the snowball is melting.
 - Use a. to find the rate at which the volume is changing at $t = 1$ min.
258. [T] The daily temperature in degrees Fahrenheit of Phoenix in the summer can be modeled by the function
 $T(x) = 94 - 10\cos\left[\frac{\pi}{12}(x - 2)\right]$,
 where x is hours after midnight. Find the rate at which the temperature is changing at 4 p.m.
259. [T] The depth (in feet) of water at a dock changes with the rise and fall of tides. The depth is modeled by the function
 $D(t) = 5\sin\left(\frac{\pi}{6}t - \frac{7\pi}{6}\right) + 8$,
 where t is the number of hours after midnight. Find the rate at which the depth is changing at 6 a.m.

3.7 Derivatives of Inverse Functions

Learning Objectives

- 3.7.1 Calculate the derivative of an inverse function.
- 3.7.2 Recognize the derivatives of the standard inverse trigonometric functions.

In this section we explore the relationship between the derivative of a function and the derivative of its inverse. For functions whose derivatives we already know, we can use this relationship to find derivatives of inverses without having to use the limit definition of the derivative. In particular, we will apply the formula for derivatives of inverse functions to trigonometric functions. This formula may also be used to extend the power rule to rational exponents.

The Derivative of an Inverse Function

We begin by considering a function and its inverse. If $f(x)$ is both invertible and differentiable, it seems reasonable that the inverse of $f(x)$ is also differentiable. Figure 3.28 shows the relationship between a function $f(x)$ and its inverse $f^{-1}(x)$. Look at the point $(a, f^{-1}(a))$ on the graph of $f^{-1}(x)$ having a tangent line with a slope of $(f^{-1})'(a) = \frac{p}{q}$. This point corresponds to a point $(f^{-1}(a), a)$ on the graph of $f(x)$ having a tangent line with a slope of $f'(f^{-1}(a)) = \frac{q}{p}$. Thus, if $f^{-1}(x)$ is differentiable at a , then it must be the case that

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

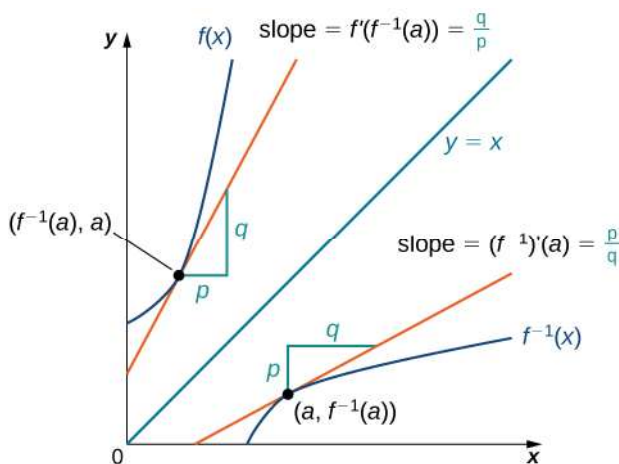


Figure 3.28 The tangent lines of a function and its inverse are related; so, too, are the derivatives of these functions.

We may also derive the formula for the derivative of the inverse by first recalling that $x = f(f^{-1}(x))$. Then by differentiating both sides of this equation (using the chain rule on the right), we obtain

$$1 = f'(f^{-1}(x)) (f^{-1})'(x).$$

Solving for $(f^{-1})'(x)$, we obtain

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}. \quad (3.19)$$

We summarize this result in the following theorem.

Theorem 3.11

Inverse Function Theorem

Let $f(x)$ be a function that is both invertible and differentiable. Let $y = f^{-1}(x)$ be the inverse of $f(x)$. For all x satisfying $f'(f^{-1}(x)) \neq 0$,

$$\frac{dy}{dx} = \frac{d}{dx}(f^{-1}(x)) = (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

Alternatively, if $y = g(x)$ is the inverse of $f(x)$, then

$$g'(x) = \frac{1}{f'(g(x))}.$$

EXAMPLE 3.60**Applying the Inverse Function Theorem**

Use the inverse function theorem to find the derivative of $g(x) = \frac{x+2}{x}$. Compare the resulting derivative to that obtained by differentiating the function directly.

✓ Solution

The inverse of $g(x) = \frac{x+2}{x}$ is $f(x) = \frac{2}{x-1}$. Since $g'(x) = \frac{1}{f'(g(x))}$, begin by finding $f'(x)$. Thus,

$$f'(x) = \frac{-2}{(x-1)^2} \text{ and } f'(g(x)) = \frac{-2}{(g(x)-1)^2} = \frac{-2}{\left(\frac{x+2}{x}-1\right)^2} = -\frac{x^2}{2}.$$

Finally,

$$g'(x) = \frac{1}{f'(g(x))} = -\frac{2}{x^2}.$$

We can verify that this is the correct derivative by applying the quotient rule to $g(x)$ to obtain

$$g'(x) = -\frac{2}{x^2}.$$

- ✓ 3.42 Use the inverse function theorem to find the derivative of $g(x) = \frac{1}{x+2}$. Compare the result obtained by differentiating $g(x)$ directly.

EXAMPLE 3.61**Applying the Inverse Function Theorem**

Use the inverse function theorem to find the derivative of $g(x) = \sqrt[3]{x}$.

✓ Solution

The function $g(x) = \sqrt[3]{x}$ is the inverse of the function $f(x) = x^3$. Since $g'(x) = \frac{1}{f'(g(x))}$, begin by finding $f'(x)$. Thus,

$$f'(x) = 3x^2 \text{ and } f'(g(x)) = 3(\sqrt[3]{x})^2 = 3x^{2/3}.$$

Finally,

$$g'(x) = \frac{1}{3x^{2/3}} = \frac{1}{3}x^{-2/3}.$$

- ✓ 3.43 Find the derivative of $g(x) = \sqrt[5]{x}$ by applying the inverse function theorem.

From the previous example, we see that we can use the inverse function theorem to extend the power rule to exponents of the form $\frac{1}{n}$, where n is a positive integer. This extension will ultimately allow us to differentiate x^q , where q is any rational number.

Theorem 3.12**Extending the Power Rule to Rational Exponents**

The power rule may be extended to rational exponents. That is, if n is a positive integer, then

$$\frac{d}{dx}(x^{1/n}) = \frac{1}{n}x^{(1/n)-1}. \quad (3.20)$$

Also, if n is a positive integer and m is an arbitrary integer, then

$$\frac{d}{dx}(x^{m/n}) = \frac{m}{n}x^{(m/n)-1}. \quad (3.21)$$

Proof

The function $g(x) = x^{1/n}$ is the inverse of the function $f(x) = x^n$. Since $g'(x) = \frac{1}{f'(g(x))}$, begin by finding $f'(x)$. Thus,

$$f'(x) = nx^{n-1} \text{ and } f'(g(x)) = n(x^{1/n})^{n-1} = nx^{(n-1)/n}.$$

Finally,

$$g'(x) = \frac{1}{nx^{(n-1)/n}} = \frac{1}{n}x^{(1-n)/n} = \frac{1}{n}x^{(1/n)-1}.$$

To differentiate $x^{m/n}$ we must rewrite it as $(x^{1/n})^m$ and apply the chain rule. Thus,

$$\frac{d}{dx}(x^{m/n}) = \frac{d}{dx}((x^{1/n})^m) = m(x^{1/n})^{m-1} \cdot \frac{1}{n}x^{(1/n)-1} = \frac{m}{n}x^{(m/n)-1}.$$

□

EXAMPLE 3.62**Applying the Power Rule to a Rational Power**

Find an equation of the line tangent to the graph of $y = x^{2/3}$ at $x = 8$.

✓ Solution

First find $\frac{dy}{dx}$ and evaluate it at $x = 8$. Since

$$\frac{dy}{dx} = \frac{2}{3}x^{-1/3} \text{ and } \left. \frac{dy}{dx} \right|_{x=8} = \frac{1}{3}$$

the slope of the tangent line to the graph at $x = 8$ is $\frac{1}{3}$.

Substituting $x = 8$ into the original function, we obtain $y = 4$. Thus, the tangent line passes through the point $(8, 4)$. Substituting into the point-slope formula for a line, we obtain the tangent line

$$y = \frac{1}{3}x + \frac{4}{3}.$$

✓ 3.44 Find the derivative of $s(t) = \sqrt{2t+1}$.

Derivatives of Inverse Trigonometric Functions

We now turn our attention to finding derivatives of inverse trigonometric functions. These derivatives will prove invaluable in the study of integration later in this text. The derivatives of inverse trigonometric functions are quite surprising in that their derivatives are actually algebraic functions. Previously, derivatives of algebraic functions have proven to be algebraic functions and derivatives of trigonometric functions have been shown to be trigonometric functions. Here, for the first time, we see that the derivative of a function need not be of the same type as the original function.

EXAMPLE 3.63**Derivative of the Inverse Sine Function**

Use the inverse function theorem to find the derivative of $g(x) = \sin^{-1}x$.

✓ Solution

Since for x in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $f(x) = \sin x$ is the inverse of $g(x) = \sin^{-1}x$, begin by finding $f'(x)$. Since

$$f'(x) = \cos x \text{ and } f'(g(x)) = \cos(\sin^{-1}x) = \sqrt{1-x^2},$$

we see that

$$g'(x) = \frac{d}{dx}(\sin^{-1} x) = \frac{1}{f'(g(x))} = \frac{1}{\sqrt{1-x^2}}.$$

Analysis

To see that $\cos(\sin^{-1} x) = \sqrt{1-x^2}$, consider the following argument. Set $\sin^{-1} x = \theta$. In this case, $\sin \theta = x$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. We begin by considering the case where $0 < \theta < \frac{\pi}{2}$. Since θ is an acute angle, we may construct a right triangle having acute angle θ , a hypotenuse of length 1 and the side opposite angle θ having length x . From the Pythagorean theorem, the side adjacent to angle θ has length $\sqrt{1-x^2}$. This triangle is shown in Figure 3.29. Using the triangle, we see that $\cos(\sin^{-1} x) = \cos \theta = \sqrt{1-x^2}$.

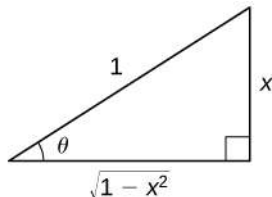


Figure 3.29 Using a right triangle having acute angle θ , a hypotenuse of length 1, and the side opposite angle θ having length x , we can see that $\cos(\sin^{-1} x) = \cos \theta = \sqrt{1-x^2}$.

In the case where $-\frac{\pi}{2} < \theta < 0$, we make the observation that $0 < -\theta < \frac{\pi}{2}$ and hence

$$\cos(\sin^{-1} x) = \cos \theta = \cos(-\theta) = \sqrt{1-x^2}.$$

Now if $\theta = \frac{\pi}{2}$ or $\theta = -\frac{\pi}{2}$, $x = 1$ or $x = -1$, and since in either case $\cos \theta = 0$ and $\sqrt{1-x^2} = 0$, we have

$$\cos(\sin^{-1} x) = \cos \theta = \sqrt{1-x^2}.$$

Finally, if $\theta = 0$, $x = 0$ and $\cos \theta = \sqrt{1-0} = 1$.

Consequently, in all cases, $\cos(\sin^{-1} x) = \sqrt{1-x^2}$.

EXAMPLE 3.64

Applying the Chain Rule to the Inverse Sine Function

Apply the chain rule to the formula derived in Example 3.61 to find the derivative of $h(x) = \sin^{-1}(g(x))$ and use this result to find the derivative of $h(x) = \sin^{-1}(2x^3)$.

Solution

Applying the chain rule to $h(x) = \sin^{-1}(g(x))$, we have

$$h'(x) = \frac{1}{\sqrt{1-(g(x))^2}} g'(x).$$

Now let $g(x) = 2x^3$, so $g'(x) = 6x^2$. Substituting into the previous result, we obtain

$$\begin{aligned} h'(x) &= \frac{1}{\sqrt{1-4x^6}} \cdot 6x^2 \\ &= \frac{6x^2}{\sqrt{1-4x^6}}. \end{aligned}$$

- ✓ 3.45 Use the inverse function theorem to find the derivative of $g(x) = \tan^{-1} x$.

The derivatives of the remaining inverse trigonometric functions may also be found by using the inverse function theorem. These formulas are provided in the following theorem.

Theorem 3.13**Derivatives of Inverse Trigonometric Functions**

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \quad (3.22)$$

$$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}} \quad (3.23)$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \quad (3.24)$$

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2} \quad (3.25)$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{|x| \sqrt{x^2-1}} \quad (3.26)$$

$$\frac{d}{dx} \csc^{-1} x = \frac{-1}{|x| \sqrt{x^2-1}} \quad (3.27)$$

EXAMPLE 3.65**Applying Differentiation Formulas to an Inverse Tangent Function**

Find the derivative of $f(x) = \tan^{-1}(x^2)$.

 Solution

Let $g(x) = x^2$, so $g'(x) = 2x$. Substituting into [Equation 3.24](#), we obtain

$$f'(x) = \frac{1}{1+(x^2)^2} \cdot (2x).$$

Simplifying, we have

$$f'(x) = \frac{2x}{1+x^4}.$$

EXAMPLE 3.66**Applying Differentiation Formulas to an Inverse Sine Function**

Find the derivative of $h(x) = x^2 \sin^{-1} x$.

 Solution

By applying the product rule, we have

$$h'(x) = 2x \sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \cdot x^2.$$

 3.46 Find the derivative of $h(x) = \cos^{-1}(3x-1)$.

EXAMPLE 3.67**Applying the Inverse Tangent Function**

The position of a particle at time t is given by $s(t) = \tan^{-1}\left(\frac{1}{t}\right)$ for $t \geq \frac{1}{2}$. Find the velocity of the particle at time $t = 1$.

 Solution

Begin by differentiating $s(t)$ in order to find $v(t)$. Thus,

$$v(t) = s'(t) = \frac{1}{1 + \left(\frac{1}{t}\right)^2} \cdot \frac{-1}{t^2}.$$

Simplifying, we have

$$v(t) = -\frac{1}{t^2 + 1}.$$

Thus, $v(1) = -\frac{1}{2}$.

-  3.47 Find an equation of the line tangent to the graph of $f(x) = \sin^{-1}x$ at $x = 0$.

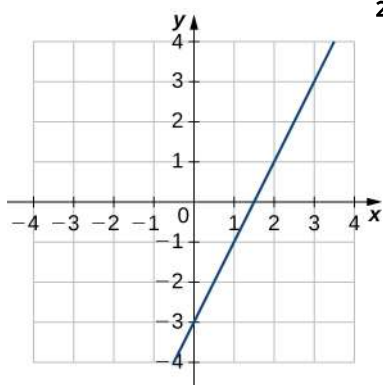


SECTION 3.7 EXERCISES

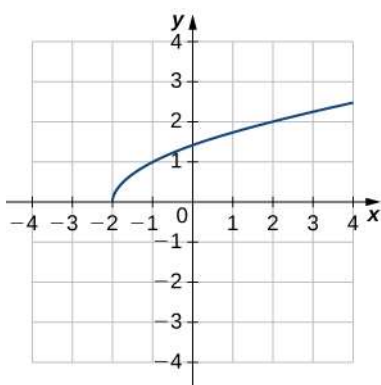
For the following exercises, use the graph of $y = f(x)$ to

- sketch the graph of $y = f^{-1}(x)$, and
- use part a. to estimate $(f^{-1})'(1)$.

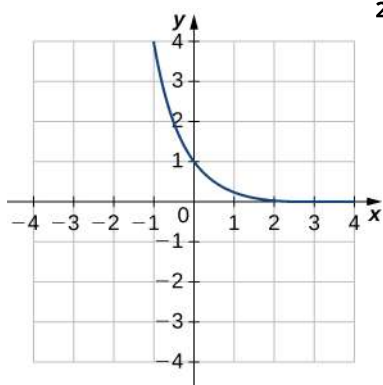
260.



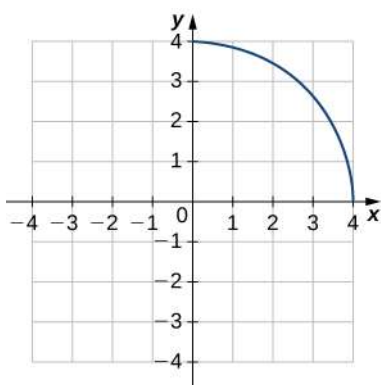
261.



262.



263.



For the following exercises, use the functions $y = f(x)$ to find

- $\frac{df}{dx}$ at $x = a$ and
- $x = f^{-1}(y)$.
- Then use part b. to find $\frac{df^{-1}}{dy}$ at $y = f(a)$.

264. $f(x) = 6x - 1, x = -2$

265. $f(x) = 2x^3 - 3, x = 1$

266. $f(x) = 9 - x^2, 0 \leq x \leq 3, x = 2$

267. $f(x) = \sin x, x = 0$

For each of the following functions, find $(f^{-1})'(a)$.

268. $f(x) = x^2 + 3x + 2, x \geq -\frac{3}{2}, a = 2$ 269. $f(x) = x^3 + 2x + 3, a = 0$ 270. $f(x) = x + \sqrt{x}, a = 2$

271. $f(x) = x - \frac{2}{x}, x < 0, a = 1$ 272. $f(x) = x + \sin x, a = 0$

273. $f(x) = \tan x + 3x^2, a = 0; 0 < x < \pi/2$

For each of the given functions $y = f(x)$,

- a. find the slope of the tangent line to its inverse function f^{-1} at the indicated point P , and
 b. find an equation of the tangent line to the graph of f^{-1} at the indicated point.

274. $f(x) = \frac{4}{1+x^2}, P(2, 1)$ 275. $f(x) = \sqrt{x-4}, P(2, 8)$ 276. $f(x) = (x^3 + 1)^4, P(16, 1)$

277. $f(x) = -x^3 - x + 2, P(-8, 2)$ 278. $f(x) = x^5 + 3x^3 - 4x - 8, P(-8, 1)$

For the following exercises, find $\frac{dy}{dx}$ for the given function.

279. $y = \sin^{-1}(x^2)$

280. $y = \cos^{-1}(\sqrt{x})$

281. $y = \sec^{-1}\left(\frac{1}{x}\right)$

282. $y = \sqrt{\csc^{-1}x}$

283. $y = (1 + \tan^{-1}x)^3$

284. $y = \cos^{-1}(2x) \cdot \sin^{-1}(2x)$

285. $y = \frac{1}{\tan^{-1}(x)}$

286. $y = \sec^{-1}(-x)$

287. $y = \cot^{-1}\sqrt{4-x^2}$

288. $y = x \cdot \csc^{-1}x$

For the following exercises, use the given values to find $(f^{-1})'(a)$.

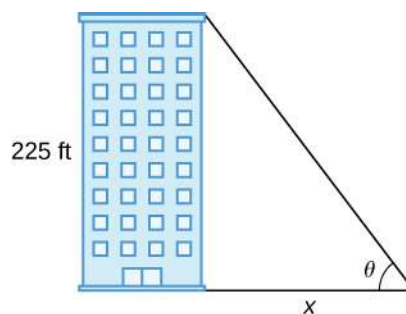
289. $f(\pi) = 0, f'(\pi) = -1, a = 0$ 290. $f(6) = 2, f'(6) = \frac{1}{3}, a = 2$ 291. $f\left(\frac{1}{3}\right) = -8, f'\left(\frac{1}{3}\right) = 2, a = -8$

292. $f(\sqrt{3}) = \frac{1}{2}, f'(\sqrt{3}) = \frac{2}{3}, a = \frac{1}{2}$ 293. $f(1) = -3, f'(1) = 10, a = -3$

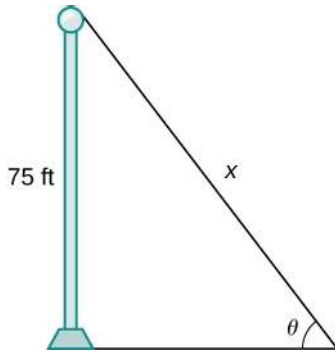
294. $f(1) = 0, f'(1) = -2, a = 0$

295. [T] The position of a moving hockey puck after t seconds is $s(t) = \tan^{-1}t$ where s is in meters.
- Find the velocity of the hockey puck at any time t .
 - Find the acceleration of the puck at any time t .
 - Evaluate a. and b. for $t = 2, 4$, and 6 seconds.
 - What conclusion can be drawn from the results in c.?

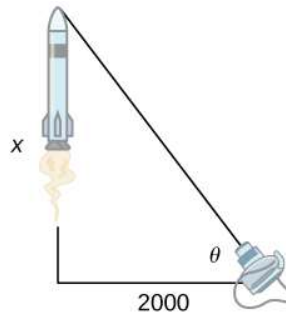
296. [T] A building that is 225 feet tall casts a shadow of various lengths x as the day goes by. An angle of elevation θ is formed by lines from the top and bottom of the building to the tip of the shadow, as seen in the following figure. Find the rate of change of the angle of elevation $\frac{d\theta}{dx}$ when $x = 272$ feet.



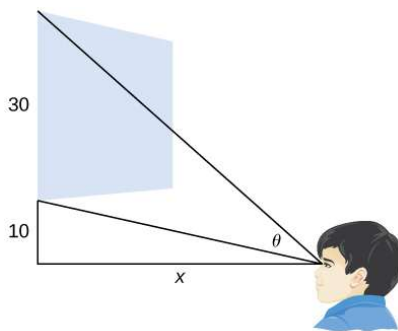
297. [T] A pole stands 75 feet tall. An angle θ is formed when wires of various lengths of x feet are attached from the ground to the top of the pole, as shown in the following figure. Find the rate of change of the angle $\frac{d\theta}{dx}$ when a wire of length 90 feet is attached.



298. [T] A television camera at ground level is 2000 feet away from the launching pad of a space rocket that is set to take off vertically, as seen in the following figure. The angle of elevation of the camera can be found by $\theta = \tan^{-1} \left(\frac{x}{2000} \right)$, where x is the height of the rocket. Find the rate of change of the angle of elevation after launch when the camera and the rocket are 5000 feet apart.



299. [T] A local movie theater with a 30-foot-high screen that is 10 feet above a person's eye level when seated has a viewing angle θ (in radians) given by $\theta = \cot^{-1} \frac{x}{40} - \cot^{-1} \frac{x}{10}$, where x is the distance in feet away from the movie screen that the person is sitting, as shown in the following figure.



- Find $\frac{d\theta}{dx}$.
- Evaluate $\frac{d\theta}{dx}$ for $x = 5, 10, 15$, and 20 .
- Interpret the results in b..
- Evaluate $\frac{d\theta}{dx}$ for $x = 25, 30, 35$, and 40 .
- Interpret the results in d. At what distance x should the person sit to maximize his or her viewing angle?

3.8 Implicit Differentiation

Learning Objectives

3.8.1 Find the derivative of a complicated function by using implicit differentiation.

3.8.2 Use implicit differentiation to determine the equation of a tangent line.

We have already studied how to find equations of tangent lines to functions and the rate of change of a function at a specific point. In all these cases we had the explicit equation for the function and differentiated these functions explicitly. Suppose instead that we want to determine the equation of a tangent line to an arbitrary curve or the rate of change of an arbitrary curve at a point. In this section, we solve these problems by finding the derivatives of functions that define y implicitly in terms of x .

Implicit Differentiation

In most discussions of math, if the dependent variable y is a function of the independent variable x , we express y in terms of x . If this is the case, we say that y is an explicit function of x . For example, when we write the equation $y = x^2 + 1$, we are defining y explicitly in terms of x . On the other hand, if the relationship between the function y and the variable x is expressed by an equation where y is not expressed entirely in terms of x , we say that the equation defines y implicitly in terms of x . For example, the equation $y - x^2 = 1$ defines the function $y = x^2 + 1$ implicitly.

Implicit differentiation allows us to find slopes of tangents to curves that are clearly not functions (they fail the vertical line test). We are using the idea that portions of y are functions that satisfy the given equation, but that y is not actually a function of x .

In general, an equation defines a function implicitly if the function satisfies that equation. An equation may define many different functions implicitly. For example, the functions

$y = \sqrt{25 - x^2}$, $y = -\sqrt{25 - x^2}$, and $y = \begin{cases} \sqrt{25 - x^2} & \text{if } -5 < x < 0 \\ -\sqrt{25 - x^2} & \text{if } 0 < x < 25 \end{cases}$, which are illustrated in [Figure 3.30](#), are just three of the many functions defined implicitly by the equation $x^2 + y^2 = 25$.

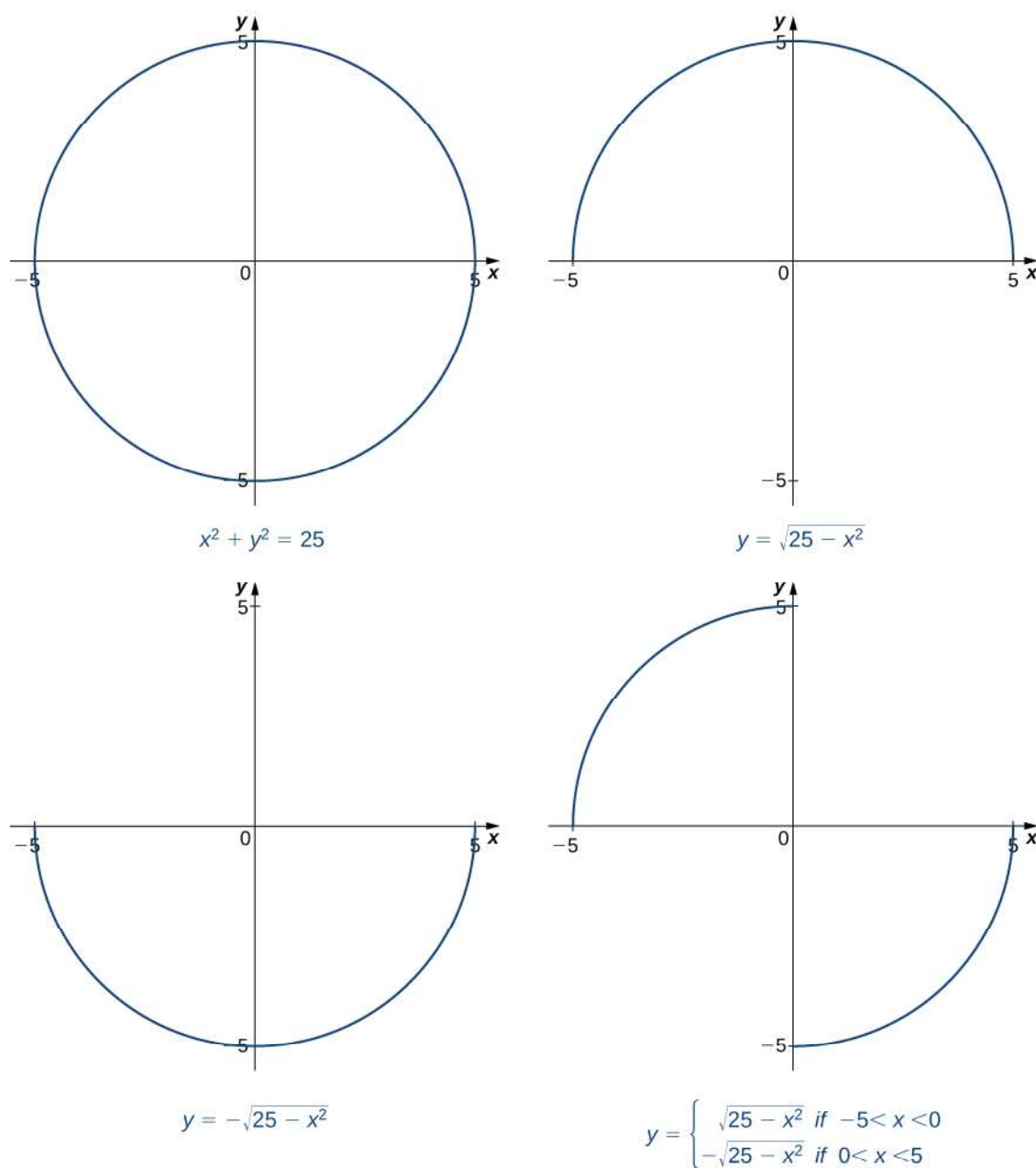


Figure 3.30 The equation $x^2 + y^2 = 25$ defines many functions implicitly.

If we want to find the slope of the line tangent to the graph of $x^2 + y^2 = 25$ at the point $(3, 4)$, we could evaluate the derivative of the function $y = \sqrt{25 - x^2}$ at $x = 3$. On the other hand, if we want the slope of the tangent line at the point $(3, -4)$, we could use the derivative of $y = -\sqrt{25 - x^2}$. However, it is not always easy to solve for a function defined implicitly by an equation. Fortunately, the technique of **implicit differentiation** allows us to find the derivative of an implicitly defined function without ever solving for the function explicitly. The process of finding $\frac{dy}{dx}$ using implicit differentiation is described in the following problem-solving strategy.

Problem-Solving Strategy

Implicit Differentiation

To perform implicit differentiation on an equation that defines a function y implicitly in terms of a variable x , use the following steps:

1. Take the derivative of both sides of the equation. Keep in mind that y is a function of x . Consequently, whereas

$\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$ because we must use the chain rule to differentiate $\sin y$ with respect to x .

2. Rewrite the equation so that all terms containing $\frac{dy}{dx}$ are on the left and all terms that do not contain $\frac{dy}{dx}$ are on the right.
3. Factor out $\frac{dy}{dx}$ on the left.
4. Solve for $\frac{dy}{dx}$ by dividing both sides of the equation by an appropriate algebraic expression.

EXAMPLE 3.68

Using Implicit Differentiation

Assuming that y is defined implicitly by the equation $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

Solution

Follow the steps in the problem-solving strategy.

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Step 1. Differentiate both sides of the equation.

Step 1.1. Use the sum rule on the left.

On the right $\frac{d}{dx}(25) = 0$.

Step 1.2. Take the derivatives, so $\frac{d}{dx}(x^2) = 2x$

and $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$.

Step 2. Keep the terms with $\frac{dy}{dx}$ on the left.

Move the remaining terms to the right.

Step 4. Divide both sides of the equation by $2y$. (Step 3 does not apply in this case.)

Analysis

Note that the resulting expression for $\frac{dy}{dx}$ is in terms of both the independent variable x and the dependent variable y .

Although in some cases it may be possible to express $\frac{dy}{dx}$ in terms of x only, it is generally not possible to do so.

EXAMPLE 3.69

Using Implicit Differentiation and the Product Rule

Assuming that y is defined implicitly by the equation $x^3 \sin y + y = 4x + 3$, find $\frac{dy}{dx}$.

✓ **Solution**

$$\begin{aligned}\frac{d}{dx}(x^3 \sin y + y) &= \frac{d}{dx}(4x + 3) \\ \frac{d}{dx}(x^3 \sin y) + \frac{d}{dx}(y) &= 4 \\ \left(\frac{d}{dx}(x^3) \cdot \sin y + \frac{d}{dx}(\sin y) \cdot x^3\right) + \frac{dy}{dx} &= 4 \\ 3x^2 \sin y + \left(\cos y \frac{dy}{dx}\right) \cdot x^3 + \frac{dy}{dx} &= 4 \\ x^3 \cos y \frac{dy}{dx} + \frac{dy}{dx} &= 4 - 3x^2 \sin y \\ \frac{dy}{dx}(x^3 \cos y + 1) &= 4 - 3x^2 \sin y \\ \frac{dy}{dx} &= \frac{4 - 3x^2 \sin y}{x^3 \cos y + 1}\end{aligned}$$

Step 1: Differentiate both sides of the equation.

Step 1.1: Apply the sum rule on the left.

On the right, $\frac{d}{dx}(4x + 3) = 4$.

Step 1.2: Use the product rule to find

$\frac{d}{dx}(x^3 \sin y)$. Observe that $\frac{d}{dx}(y) = \frac{dy}{dx}$.

Step 1.3: We know $\frac{d}{dx}(x^3) = 3x^2$. Use the chain rule to obtain $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$.

Step 2: Keep all terms containing $\frac{dy}{dx}$ on the left. Move all other terms to the right.

Step 3: Factor out $\frac{dy}{dx}$ on the left.

Step 4: Solve for $\frac{dy}{dx}$ by dividing both sides of the equation by $x^3 \cos y + 1$.

EXAMPLE 3.70

Using Implicit Differentiation to Find a Second Derivative

Find $\frac{d^2 y}{dx^2}$ if $x^2 + y^2 = 25$.

✓ **Solution**

In [Example 3.68](#), we showed that $\frac{dy}{dx} = -\frac{x}{y}$. We can take the derivative of both sides of this equation to find $\frac{d^2 y}{dx^2}$.

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{d}{dx}\left(-\frac{x}{y}\right) && \text{Differentiate both sides of } \frac{dy}{dx} = -\frac{x}{y}. \\ &= -\frac{(1 \cdot y - x \frac{dy}{dx})}{y^2} && \text{Use the quotient rule to find } \frac{d}{dx}\left(-\frac{x}{y}\right). \\ &= \frac{-y + x \frac{dy}{dx}}{y^2} && \text{Simplify.} \\ &= \frac{-y + x\left(-\frac{x}{y}\right)}{y^2} && \text{Substitute } \frac{dy}{dx} = -\frac{x}{y}. \\ &= \frac{-y^2 - x^2}{y^3} && \text{Simplify.}\end{aligned}$$

At this point we have found an expression for $\frac{d^2 y}{dx^2}$. If we choose, we can simplify the expression further by recalling that $x^2 + y^2 = 25$ and making this substitution in the numerator to obtain $\frac{d^2 y}{dx^2} = -\frac{25}{y^3}$.

✓ 3.48 Find $\frac{dy}{dx}$ for y defined implicitly by the equation $4x^5 + \tan y = y^2 + 5x$.

Finding Tangent Lines Implicitly

Now that we have seen the technique of implicit differentiation, we can apply it to the problem of finding equations of tangent lines to curves described by equations.

EXAMPLE 3.71**Finding a Tangent Line to a Circle**

Find an equation of the line tangent to the curve $x^2 + y^2 = 25$ at the point $(3, -4)$.

✓ Solution

Although we could find this equation without using implicit differentiation, using that method makes it much easier. In [Example 3.68](#), we found $\frac{dy}{dx} = -\frac{x}{y}$.

The slope of the tangent line is found by substituting $(3, -4)$ into this expression. Consequently, the slope of the tangent line is $\left. \frac{dy}{dx} \right|_{(3, -4)} = -\frac{3}{-4} = \frac{3}{4}$.

Using the point $(3, -4)$ and the slope $\frac{3}{4}$ in the point-slope equation of the line, we obtain the equation $y = \frac{3}{4}x - \frac{25}{4}$ ([Figure 3.31](#)).

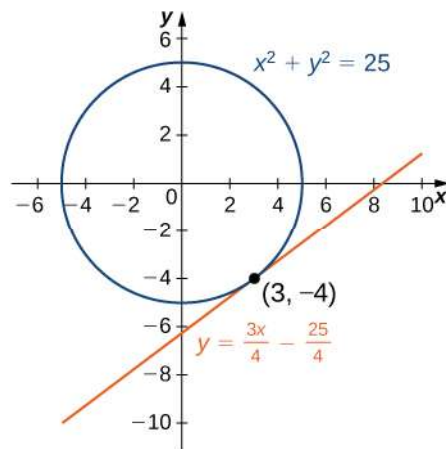


Figure 3.31 The line $y = \frac{3}{4}x - \frac{25}{4}$ is tangent to $x^2 + y^2 = 25$ at the point $(3, -4)$.

EXAMPLE 3.72**Finding the Equation of the Tangent Line to a Curve**

Find an equation of the line tangent to the graph of $y^3 + x^3 - 3xy = 0$ at the point $(\frac{3}{2}, \frac{3}{2})$ ([Figure 3.32](#)). This curve is known as the folium (or leaf) of Descartes.

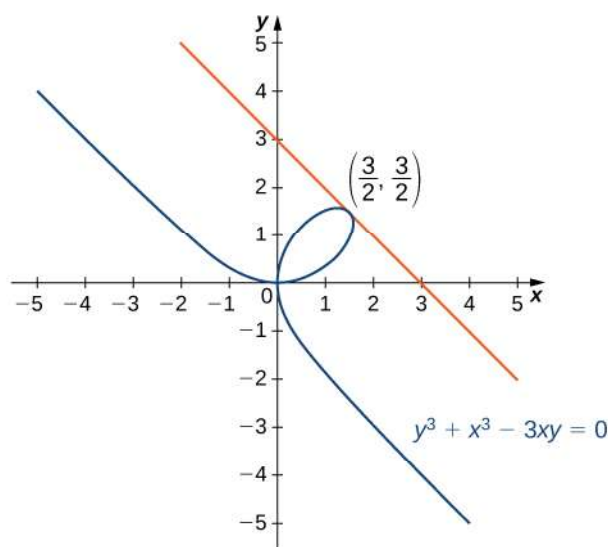


Figure 3.32 Finding the tangent line to the folium of Descartes at $(\frac{3}{2}, \frac{3}{2})$.

✓ Solution

Begin by finding $\frac{dy}{dx}$.

$$\begin{aligned}\frac{d}{dx}(y^3 + x^3 - 3xy) &= \frac{d}{dx}(0) \\ 3y^2 \frac{dy}{dx} + 3x^2 - \left(3y + \frac{dy}{dx}3x\right) &= 0 \\ \frac{dy}{dx} &= \frac{3y - 3x^2}{3y^2 - 3x}.\end{aligned}$$

Next, substitute $(\frac{3}{2}, \frac{3}{2})$ into $\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x}$ to find the slope of the tangent line:

$$\left. \frac{dy}{dx} \right|_{\left(\frac{3}{2}, \frac{3}{2}\right)} = -1.$$

Finally, substitute into the point-slope equation of the line to obtain

$$y = -x + 3.$$

EXAMPLE 3.73

Applying Implicit Differentiation

In a simple video game, a rocket travels in an elliptical orbit whose path is described by the equation $4x^2 + 25y^2 = 100$. The rocket can fire missiles along lines tangent to its path. The object of the game is to destroy an incoming asteroid traveling along the positive x -axis toward $(0, 0)$. If the rocket fires a missile when it is located at $(3, \frac{8}{5})$, where will it intersect the x -axis?

✓ Solution

To solve this problem, we must determine where the line tangent to the graph of

$4x^2 + 25y^2 = 100$ at $(3, \frac{8}{5})$ intersects the x -axis. Begin by finding $\frac{dy}{dx}$ implicitly.

Differentiating, we have

$$8x + 50y \frac{dy}{dx} = 0.$$

Solving for $\frac{dy}{dx}$, we have

$$\frac{dy}{dx} = -\frac{4x}{25y}.$$

The slope of the tangent line is $\left. \frac{dy}{dx} \right|_{\left(3, \frac{8}{5}\right)} = -\frac{3}{10}$. The equation of the tangent line is $y = -\frac{3}{10}x + \frac{5}{2}$. To determine where the line intersects the x-axis, solve $0 = -\frac{3}{10}x + \frac{5}{2}$. The solution is $x = \frac{25}{3}$. The missile intersects the x-axis at the point $\left(\frac{25}{3}, 0\right)$.

- ✓ 3.49 Find an equation of the line tangent to the hyperbola $x^2 - y^2 = 16$ at the point $(5, 3)$.



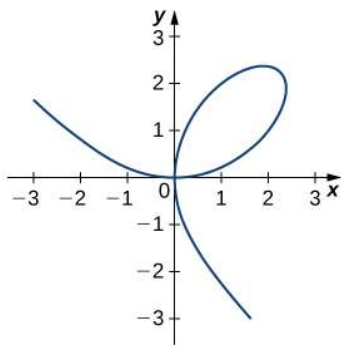
SECTION 3.8 EXERCISES

For the following exercises, use implicit differentiation to find $\frac{dy}{dx}$.

300. $x^2 - y^2 = 4$ 301. $6x^2 + 3y^2 = 12$ 302. $x^2y = y - 7$
 303. $3x^3 + 9xy^2 = 5x^3$ 304. $xy - \cos(xy) = 1$ 305. $y\sqrt{x+4} = xy + 8$
 306. $-xy - 2 = \frac{x}{7}$ 307. $y \sin(xy) = y^2 - 2$ 308. $(xy)^2 + 3x = y^2$
 309. $x^3y + xy^3 = -8$

For the following exercises, find an equation of the tangent line to the graph of the given equation at the indicated point. Use a calculator or computer software to graph the function and the tangent line.

310. [T] $x^4y - xy^3 = -2$, $(-1, -1)$ 311. [T] $x^2y^2 + 5xy = 14$, $(2, 1)$ 312. [T] $\tan(xy) = y$, $\left(\frac{\pi}{4}, 1\right)$
 313. [T] $xy^2 + \sin(\pi y) - 2x^2 = 10$, $(2, -3)$ 314. [T] $\frac{x}{y} + 5x - 7 = -\frac{3}{4}y$, $(1, 2)$ 315. [T] $xy + \sin(x) = 1$, $\left(\frac{\pi}{2}, 0\right)$
 316. [T] The graph of a folium of Descartes with equation $2x^3 + 2y^3 - 9xy = 0$ is given in the following graph.



- a. Find an equation of the tangent line at the point $(2, 1)$. Graph the tangent line along with the folium.
 b. Find an equation of the normal line to the tangent line in a. at the point $(2, 1)$.

318. Find all points on the graph of $y^3 - 27y = x^2 - 90$ at which the tangent line is vertical.

- 319.** For the equation $x^2 + xy + y^2 = 7$,
- Find the x -intercept(s).
 - Find the slope of the tangent line(s) at the x -intercept(s).
 - What does the value(s) in b. indicate about the tangent line(s)?
- 320.** Find an equation of the tangent line to the graph of the equation $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{6}$ at the point $(0, \frac{1}{2})$.
- 321.** Find an equation of the tangent line to the graph of the equation $\tan^{-1}(x + y) = x^2 + \frac{\pi}{4}$ at the point $(0, 1)$.
- 322.** Find y' and y'' for $x^2 + 6xy - 2y^2 = 3$.
- 323. [T]** The number of cell phones produced when x dollars is spent on labor and y dollars is spent on capital invested by a manufacturer can be modeled by the equation $60x^{3/4}y^{1/4} = 3240$.
- Find $\frac{dy}{dx}$ and evaluate at the point $(81, 16)$.
 - Interpret the result of a.
- 324. [T]** The number of cars produced when x dollars is spent on labor and y dollars is spent on capital invested by a manufacturer can be modeled by the equation $30x^{1/3}y^{2/3} = 360$. (Both x and y are measured in thousands of dollars.)
- Find $\frac{dy}{dx}$ and evaluate at the point $(27, 8)$.
 - Interpret the result of a.
- 325.** The volume of a right circular cone of radius x and height y is given by $V = \frac{1}{3}\pi x^2 y$. Suppose that the volume of the cone is a constant. Find $\frac{dy}{dx}$ when $x = 4$ and $y = 16$.

For the following exercises, consider a closed rectangular box with a square base with side x and height y .

- 326.** Find an equation for the surface area of the rectangular box, $S(x, y)$.
- 327.** If the surface area of the rectangular box is 78 square feet, find $\frac{dy}{dx}$ when $x = 3$ feet and $y = 5$ feet.

For the following exercises, use implicit differentiation to determine y' . Does the answer agree with the formulas we have previously determined?

- 328.** $x = \sin y$ **329.** $x = \cos y$ **330.** $x = \tan y$

3.9 Derivatives of Exponential and Logarithmic Functions

Learning Objectives

- 3.9.1** Find the derivative of exponential functions.
3.9.2 Find the derivative of logarithmic functions.
3.9.3 Use logarithmic differentiation to determine the derivative of a function.

So far, we have learned how to differentiate a variety of functions, including trigonometric, inverse, and implicit functions. In this section, we explore derivatives of exponential and logarithmic functions. As we discussed in [Introduction to Functions and Graphs](#), exponential functions play an important role in modeling population growth and the decay of radioactive materials. Logarithmic functions can help rescale large quantities and are particularly helpful for rewriting complicated expressions.

Derivative of the Exponential Function

Just as when we found the derivatives of other functions, we can find the derivatives of exponential and logarithmic functions using formulas. As we develop these formulas, we need to make certain basic assumptions. The proofs that these assumptions hold are beyond the scope of this course.

First of all, we begin with the assumption that the function $B(x) = b^x$, $b > 0$, is defined for every real number and is continuous. In previous courses, the values of exponential functions for all rational numbers were defined—beginning with the definition of b^n , where n is a positive integer—as the product of b multiplied by itself n times. Later, we defined $b^0 = 1$, $b^{-n} = \frac{1}{b^n}$, for a positive integer n , and $b^{s/t} = (\sqrt[t]{b})^s$ for positive integers s and t . These definitions leave open the question of the value of b^r where r is an arbitrary real number. By assuming the *continuity* of $B(x) = b^x$, $b > 0$, we may interpret b^r as $\lim_{x \rightarrow r} b^x$ where the values of x as we take the limit are rational. For example, we may view 4^π as the number satisfying

$$4^3 < 4^\pi < 4^4, 4^{3.1} < 4^\pi < 4^{3.2}, 4^{3.14} < 4^\pi < 4^{3.15}, \\ 4^{3.141} < 4^\pi < 4^{3.142}, 4^{3.1415} < 4^\pi < 4^{3.1416}, \dots$$

As we see in the following table, $4^\pi \approx 77.88$.

x	4^x	x	4^x
3	64	3.141593	77.8802710486
3.1	73.5166947198	3.1416	77.8810268071
3.14	77.7084726013	3.142	77.9242251944
3.141	77.8162741237	3.15	78.7932424541
3.1415	77.8702309526	3.2	84.4485062895
3.14159	77.8799471543	4	256

Table 3.6 Approximating a Value of 4^π

We also assume that for $B(x) = b^x$, $b > 0$, the value $B'(0)$ of the derivative exists. In this section, we show that by making this one additional assumption, it is possible to prove that the function $B(x)$ is differentiable everywhere.

We make one final assumption: that there is a unique value of $b > 0$ for which $B'(0) = 1$. We define e to be this unique value, as we did in [Introduction to Functions and Graphs](#). [Figure 3.33](#) provides graphs of the functions $y = 2^x$, $y = 3^x$, $y = 2.7^x$, and $y = 2.8^x$. A visual estimate of the slopes of the tangent lines to these functions at 0 provides evidence that the value of e lies somewhere between 2.7 and 2.8. The function $E(x) = e^x$ is called the **natural exponential function**. Its inverse, $L(x) = \log_e x = \ln x$ is called the **natural logarithmic function**.

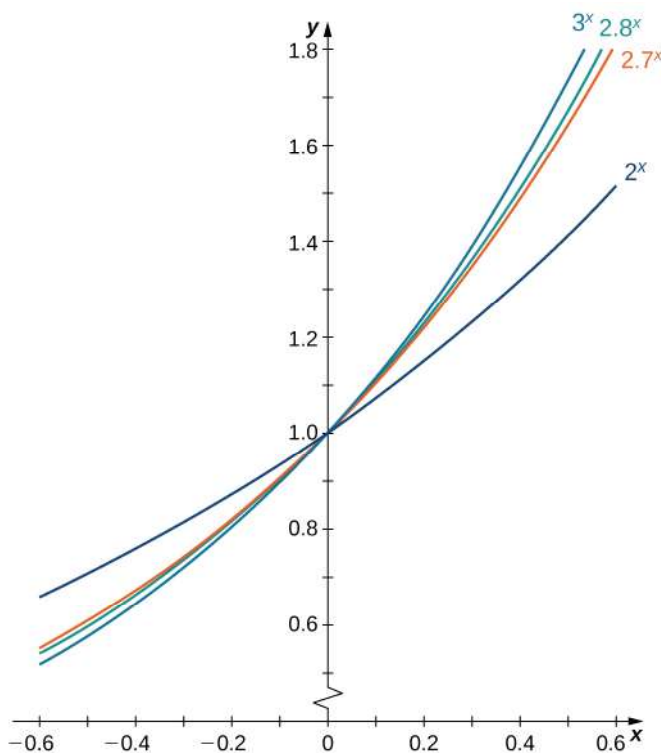


Figure 3.33 The graph of $E(x) = e^x$ is between $y = 2^x$ and $y = 3^x$.

For a better estimate of e , we may construct a table of estimates of $B'(0)$ for functions of the form $B(x) = b^x$. Before doing this, recall that

$$B'(0) = \lim_{x \rightarrow 0} \frac{b^x - b^0}{x - 0} = \lim_{x \rightarrow 0} \frac{b^x - 1}{x} \approx \frac{b^x - 1}{x}$$

for values of x very close to zero. For our estimates, we choose $x = 0.00001$ and $x = -0.00001$ to obtain the estimate

$$\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}.$$

See the following table.

b	$\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}$	b	$\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}$
2	$0.693145 < B'(0) < 0.69315$	2.7183	$1.000002 < B'(0) < 1.000012$
2.7	$0.993247 < B'(0) < 0.993257$	2.719	$1.000259 < B'(0) < 1.000269$
2.71	$0.996944 < B'(0) < 0.996954$	2.72	$1.000627 < B'(0) < 1.000637$
2.718	$0.999891 < B'(0) < 0.999901$	2.8	$1.029614 < B'(0) < 1.029625$
2.7182	$0.999965 < B'(0) < 0.999975$	3	$1.098606 < B'(0) < 1.098618$

Table 3.7 Estimating a Value of e

The evidence from the table suggests that $2.7182 < e < 2.7183$.

The graph of $E(x) = e^x$ together with the line $y = x + 1$ are shown in [Figure 3.34](#). This line is tangent to the graph of $E(x) = e^x$ at $x = 0$.

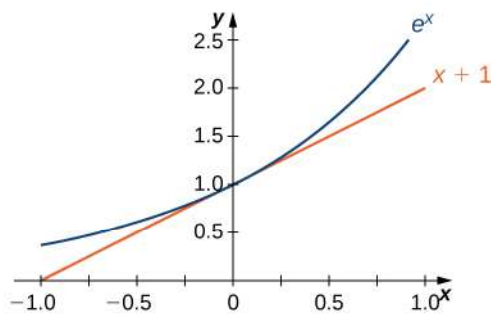


Figure 3.34 The tangent line to $E(x) = e^x$ at $x = 0$ has slope 1.

Now that we have laid out our basic assumptions, we begin our investigation by exploring the derivative of $B(x) = b^x$, $b > 0$. Recall that we have assumed that $B'(0)$ exists. By applying the limit definition to the derivative we conclude that

$$B'(0) = \lim_{h \rightarrow 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \rightarrow 0} \frac{b^h - 1}{h}. \quad (3.28)$$

Turning to $B'(x)$, we obtain the following.

$$\begin{aligned} B'(x) &= \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} && \text{Apply the limit definition of the derivative.} \\ &= \lim_{h \rightarrow 0} \frac{b^x b^h - b^x}{h} && \text{Note that } b^{x+h} = b^x b^h. \\ &= \lim_{h \rightarrow 0} \frac{b^x (b^h - 1)}{h} && \text{Factor out } b^x. \\ &= b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h} && \text{Apply a property of limits.} \\ &= b^x B'(0) && \text{Use } B'(0) = \lim_{h \rightarrow 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \rightarrow 0} \frac{b^h - 1}{h}. \end{aligned}$$

We see that on the basis of the assumption that $B(x) = b^x$ is differentiable at 0, $B(x)$ is not only differentiable everywhere, but its derivative is

$$B'(x) = b^x B'(0). \quad (3.29)$$

For $E(x) = e^x$, $E'(0) = 1$. Thus, we have $E'(x) = e^x$. (The value of $B'(0)$ for an arbitrary function of the form $B(x) = b^x$, $b > 0$, will be derived later.)

Theorem 3.14

Derivative of the Natural Exponential Function

Let $E(x) = e^x$ be the natural exponential function. Then

$$E'(x) = e^x.$$

In general,

$$\frac{d}{dx} (e^{g(x)}) = e^{g(x)} g'(x).$$

EXAMPLE 3.74

Derivative of an Exponential Function

Find the derivative of $f(x) = e^{\tan(2x)}$.

Solution

Using the derivative formula and the chain rule,

$$\begin{aligned} f'(x) &= e^{\tan(2x)} \frac{d}{dx}(\tan(2x)) \\ &= e^{\tan(2x)} \sec^2(2x) \cdot 2. \end{aligned}$$

EXAMPLE 3.75**Combining Differentiation Rules**

Find the derivative of $y = \frac{e^{x^2}}{x}$.

✓ Solution

Use the derivative of the natural exponential function, the quotient rule, and the chain rule.

$$\begin{aligned} y' &= \frac{(e^{x^2} \cdot 2)x - x \cdot e^{x^2}}{x^2} && \text{Apply the quotient rule.} \\ &= \frac{e^{x^2}(2x^2 - 1)}{x^2} && \text{Simplify.} \end{aligned}$$

✓ 3.50 Find the derivative of $h(x) = xe^{2x}$.

EXAMPLE 3.76**Applying the Natural Exponential Function**

A colony of mosquitoes has an initial population of 1000. After t days, the population is given by $A(t) = 1000e^{0.3t}$. Show that the ratio of the rate of change of the population, $A'(t)$, to the population, $A(t)$ is constant.

✓ Solution

First find $A'(t)$. By using the chain rule, we have $A'(t) = 300e^{0.3t}$. Thus, the ratio of the rate of change of the population to the population is given by

$$\frac{A'(t)}{A(t)} = \frac{300e^{0.3t}}{1000e^{0.3t}} = 0.3.$$

The ratio of the rate of change of the population to the population is the constant 0.3.

✓ 3.51 If $A(t) = 1000e^{0.3t}$ describes the mosquito population after t days, as in the preceding example, what is the rate of change of $A(t)$ after 4 days?

Derivative of the Logarithmic Function

Now that we have the derivative of the natural exponential function, we can use implicit differentiation to find the derivative of its inverse, the natural logarithmic function.

Theorem 3.15**The Derivative of the Natural Logarithmic Function**

If $x > 0$ and $y = \ln x$, then

$$\frac{dy}{dx} = \frac{1}{x}. \quad (3.30)$$

More generally, let $g(x)$ be a differentiable function. For all values of x for which $g(x) > 0$, the derivative of $h(x) = \ln(g(x))$ is given by

$$h'(x) = \frac{1}{g(x)} g'(x). \quad (3.31)$$

Proof

If $x > 0$ and $y = \ln x$, then $e^y = x$. Differentiating both sides of this equation results in the equation

$$e^y \frac{dy}{dx} = 1.$$

Solving for $\frac{dy}{dx}$ yields

$$\frac{dy}{dx} = \frac{1}{e^y}.$$

Finally, we substitute $x = e^y$ to obtain

$$\frac{dy}{dx} = \frac{1}{x}.$$

We may also derive this result by applying the inverse function theorem, as follows. Since $y = g(x) = \ln x$ is the inverse of $f(x) = e^x$, by applying the inverse function theorem we have

$$\frac{dy}{dx} = \frac{1}{f'(g(x))} = \frac{1}{e^{\ln x}} = \frac{1}{x}.$$

Using this result and applying the chain rule to $h(x) = \ln(g(x))$ yields

$$h'(x) = \frac{1}{g(x)} g'(x).$$

□

The graph of $y = \ln x$ and its derivative $\frac{dy}{dx} = \frac{1}{x}$ are shown in [Figure 3.35](#).

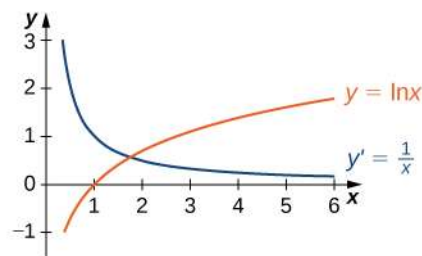


Figure 3.35 The function $y = \ln x$ is increasing on $(0, +\infty)$. Its derivative $y' = \frac{1}{x}$ is greater than zero on $(0, +\infty)$.

EXAMPLE 3.77**Taking a Derivative of a Natural Logarithm**

Find the derivative of $f(x) = \ln(x^3 + 3x - 4)$.

✓ Solution

Use [Equation 3.31](#) directly.

$$\begin{aligned} f'(x) &= \frac{1}{x^3 + 3x - 4} \cdot (3x^2 + 3) && \text{Use } g(x) = x^3 + 3x - 4 \text{ in } h'(x) = \frac{1}{g(x)} g'(x). \\ &= \frac{3x^2 + 3}{x^3 + 3x - 4} && \text{Rewrite.} \end{aligned}$$

EXAMPLE 3.78**Using Properties of Logarithms in a Derivative**

Find the derivative of $f(x) = \ln \left(\frac{x^2 \sin x}{2x+1} \right)$.

✓ Solution

At first glance, taking this derivative appears rather complicated. However, by using the properties of logarithms prior to finding the derivative, we can make the problem much simpler.

$$f(x) = \ln \left(\frac{x^2 \sin x}{2x+1} \right) = 2 \ln x + \ln(\sin x) - \ln(2x+1) \quad \text{Apply properties of logarithms.}$$

$$f'(x) = \frac{2}{x} + \cot x - \frac{2}{2x+1} \quad \text{Apply sum rule and } h'(x) = \frac{1}{g(x)} g'(x).$$

✓ 3.52 Differentiate: $f(x) = \ln(3x+2)^5$.

Now that we can differentiate the natural logarithmic function, we can use this result to find the derivatives of $y = \log_b x$ and $y = b^x$ for $b > 0$, $b \neq 1$.

Theorem 3.16**Derivatives of General Exponential and Logarithmic Functions**

Let $b > 0$, $b \neq 1$, and let $g(x)$ be a differentiable function.

i. If $y = \log_b x$, then

$$\frac{dy}{dx} = \frac{1}{x \ln b}. \quad (3.32)$$

More generally, if $h(x) = \log_b(g(x))$, then for all values of x for which $g(x) > 0$,

$$h'(x) = \frac{g'(x)}{g(x) \ln b}. \quad (3.33)$$

ii. If $y = b^x$, then

$$\frac{dy}{dx} = b^x \ln b. \quad (3.34)$$

More generally, if $h(x) = b^{g(x)}$, then

$$h'(x) = b^{g(x)} g'(x) \ln b. \quad (3.35)$$

Proof

If $y = \log_b x$, then $b^y = x$. It follows that $\ln(b^y) = \ln x$. Thus $y \ln b = \ln x$. Solving for y , we have $y = \frac{\ln x}{\ln b}$. Differentiating and keeping in mind that $\ln b$ is a constant, we see that

$$\frac{dy}{dx} = \frac{1}{x \ln b}.$$

The derivative in [Equation 3.32](#) now follows from the chain rule.

If $y = b^x$, then $\ln y = x \ln b$. Using implicit differentiation, again keeping in mind that $\ln b$ is constant, it follows that $\frac{1}{y} \frac{dy}{dx} = \ln b$. Solving for $\frac{dy}{dx}$ and substituting $y = b^x$, we see that

$$\frac{dy}{dx} = y \ln b = b^x \ln b.$$

The more general derivative ([Equation 3.35](#)) follows from the chain rule.

□

EXAMPLE 3.79**Applying Derivative Formulas**

Find the derivative of $h(x) = \frac{3^x}{3^x + 2}$.

✓ Solution

Use the quotient rule and [Derivatives of General Exponential and Logarithmic Functions](#).

$$\begin{aligned} h'(x) &= \frac{3^x \ln 3 (3^x + 2) - 3^x \ln 3 (3^x)}{(3^x + 2)^2} && \text{Apply the quotient rule.} \\ &= \frac{2 \cdot 3^x \ln 3}{(3^x + 2)^2} && \text{Simplify.} \end{aligned}$$

EXAMPLE 3.80**Finding the Slope of a Tangent Line**

Find the slope of the line tangent to the graph of $y = \log_2(3x + 1)$ at $x = 1$.

✓ Solution

To find the slope, we must evaluate $\frac{dy}{dx}$ at $x = 1$. Using [Equation 3.33](#), we see that

$$\frac{dy}{dx} = \frac{3}{(3x + 1) \ln 2}.$$

By evaluating the derivative at $x = 1$, we see that the tangent line has slope

$$\left. \frac{dy}{dx} \right|_{x=1} = \frac{3}{4 \ln 2} = \frac{3}{\ln 16}.$$

✓ 3.53 Find the slope for the line tangent to $y = 3^x$ at $x = 2$.

Logarithmic Differentiation

At this point, we can take derivatives of functions of the form $y = (g(x))^n$ for certain values of n , as well as functions of the form $y = b^{g(x)}$, where $b > 0$ and $b \neq 1$. Unfortunately, we still do not know the derivatives of functions such as $y = x^x$ or $y = x^\pi$. These functions require a technique called **logarithmic differentiation**, which allows us to differentiate any function of the form $h(x) = g(x)^{f(x)}$. It can also be used to convert a very complex differentiation problem into a simpler one, such as finding the derivative of $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$. We outline this technique in the following problem-solving strategy.

Problem-Solving Strategy**Using Logarithmic Differentiation**

1. To differentiate $y = h(x)$ using logarithmic differentiation, take the natural logarithm of both sides of the equation to obtain $\ln y = \ln(h(x))$.
2. Use properties of logarithms to expand $\ln(h(x))$ as much as possible.
3. Differentiate both sides of the equation. On the left we will have $\frac{1}{y} \frac{dy}{dx}$.
4. Multiply both sides of the equation by y to solve for $\frac{dy}{dx}$.
5. Replace y by $h(x)$.

EXAMPLE 3.81**Using Logarithmic Differentiation**Find the derivative of $y = (2x^4 + 1)^{\tan x}$. **Solution**

Use logarithmic differentiation to find this derivative.

$$\ln y = \ln(2x^4 + 1)^{\tan x}$$

$$\ln y = \tan x \ln(2x^4 + 1)$$

$$\frac{1}{y} \frac{dy}{dx} = \sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x$$

$$\frac{dy}{dx} = y \cdot \left(\sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x \right)$$

$$\frac{dy}{dx} = (2x^4 + 1)^{\tan x} \left(\sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x \right)$$

Step 1. Take the natural logarithm of both sides.

Step 2. Expand using properties of logarithms.

Step 3. Differentiate both sides. Use the product rule on the right.

Step 4. Multiply by y on both sides.Step 5. Substitute $y = (2x^4 + 1)^{\tan x}$.**EXAMPLE 3.82****Using Logarithmic Differentiation**Find the derivative of $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$. **Solution**

This problem really makes use of the properties of logarithms and the differentiation rules given in this chapter.

$$\ln y = \ln \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$$

$$\ln y = \ln x + \frac{1}{2} \ln(2x+1) - x \ln e - 3 \ln \sin x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \frac{\cos x}{\sin x}$$

$$\frac{dy}{dx} = y \left(\frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \cot x \right)$$

$$\frac{dy}{dx} = \frac{x\sqrt{2x+1}}{e^x \sin^3 x} \left(\frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \cot x \right)$$

Step 1. Take the natural logarithm of both sides.

Step 2. Expand using properties of logarithms.

Step 3. Differentiate both sides.

Step 4. Multiply by y on both sides.Step 5. Substitute $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$.**EXAMPLE 3.83****Extending the Power Rule**Find the derivative of $y = x^r$ where r is an arbitrary real number. **Solution**The process is the same as in [Example 3.82](#), though with fewer complications.

$$\ln y = \ln x^r$$

$$\ln y = r \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = r \frac{1}{x}$$

$$\frac{dy}{dx} = y \frac{r}{x}$$

$$\frac{dy}{dx} = x^r \frac{r}{x}$$

$$\frac{dy}{dx} = rx^{r-1}$$

Step 1. Take the natural logarithm of both sides.

Step 2. Expand using properties of logarithms.

Step 3. Differentiate both sides.

Step 4. Multiply by y on both sides.Step 5. Substitute $y = x^r$.

Simplify.

 3.54 Use logarithmic differentiation to find the derivative of $y = x^x$.

- ✓ 3.55 Find the derivative of $y = (\tan x)^\pi$.



SECTION 3.9 EXERCISES

For the following exercises, find $f'(x)$ for each function.

331. $f(x) = x^2 e^x$ 332. $f(x) = \frac{e^{-x}}{x}$ 333. $f(x) = e^{x^3 \ln x}$
 334. $f(x) = \sqrt{e^{2x} + 2x}$ 335. $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ 336. $f(x) = \frac{10^x}{\ln 10}$
 337. $f(x) = 2^{4x} + 4x^2$ 338. $f(x) = 3^{\sin 3x}$ 339. $f(x) = x^\pi \cdot \pi^x$
 340. $f(x) = \ln(4x^3 + x)$ 341. $f(x) = \ln \sqrt{5x - 7}$ 342. $f(x) = x^2 \ln 9x$
 343. $f(x) = \log(\sec x)$ 344. $f(x) = \log_7(6x^4 + 3)^5$ 345. $f(x) = 2^x \cdot \log_3 7^{x^2 - 4}$

For the following exercises, use logarithmic differentiation to find $\frac{dy}{dx}$.

346. $y = x\sqrt{x}$ 347. $y = (\sin 2x)^{4x}$ 348. $y = (\ln x)^{\ln x}$
 349. $y = x^{\log_2 x}$ 350. $y = (x^2 - 1)^{\ln x}$ 351. $y = x^{\cot x}$
 352. $y = \frac{x+11}{\sqrt[3]{x^2-4}}$ 353. $y = x^{-1/2}(x^2 + 3)^{2/3}(3x - 4)^4$ 354. [T] Find an equation of the tangent line to the graph of $f(x) = 4xe^{(x^2-1)}$ at the point where $x = -1$. Graph both the function and the tangent line.
 355. [T] Find an equation of the line that is normal to the graph of $f(x) = x \cdot 5^x$ at the point where $x = 1$. Graph both the function and the normal line.
 356. [T] Find an equation of the tangent line to the graph of $x^3 - x \ln y + y^3 = 2x + 5$ at the point $(2, 1)$. (Hint: Use implicit differentiation to find $\frac{dy}{dx}$.) Graph both the curve and the tangent line.
 357. Consider the function $y = x^{1/x}$ for $x > 0$.
 a. Determine the points on the graph where the tangent line is horizontal.
 b. Determine the points on the graph where $y' > 0$ and those where $y' < 0$.

- 358.** The formula $I(t) = \frac{\sin t}{e^t}$ is the formula for a decaying alternating current.

a. Complete the following table with the appropriate values.

t	$\frac{\sin t}{e^t}$
0	(i)
$\frac{\pi}{2}$	(ii)
π	(iii)
$\frac{3\pi}{2}$	(iv)
2π	(v)
$\frac{5\pi}{2}$	(vi)
3π	(vii)
$\frac{7\pi}{2}$	(viii)
4π	(ix)

b. Using only the values in the table, determine where the tangent line to the graph of $I(t)$ is horizontal.

- 359. [T]** The population of Toledo, Ohio, in 2000 was approximately 500,000. Assume the population is increasing at a rate of 5% per year.
- Write the exponential function that relates the total population as a function of t .
 - Use a. to determine the rate at which the population is increasing in t years.
 - Use b. to determine the rate at which the population is increasing in 10 years.

- 360. [T]** An isotope of the element erbium has a half-life of approximately 12 hours. Initially there are 9 grams of the isotope present.
- Write the exponential function that relates the amount of substance remaining as a function of t , measured in hours.
 - Use a. to determine the rate at which the substance is decaying in t hours.
 - Use b. to determine the rate of decay at $t = 4$ hours.

- 361. [T]** The number of cases of influenza in New York City from the beginning of 1960 to the beginning of 1964 is modeled by the function

$$N(t) = 5.3e^{0.093t^2 - 0.87t}, \quad (0 \leq t \leq 4),$$

where $N(t)$ gives the number of cases (in thousands) and t is measured in years, with $t = 0$ corresponding to the beginning of 1960.

- Show work that evaluates $N(0)$ and $N(4)$. Briefly describe what these values indicate about the disease in New York City.
- Show work that evaluates $N'(0)$ and $N'(3)$. Briefly describe what these values indicate about the disease in New York City.

- 362. [T]** The *relative rate of change* of a differentiable function $y = f(x)$ is given by $\frac{100 \cdot f'(x)}{f(x)}\%$. One model

for population growth is a Gompertz growth function, given by

$$P(x) = ae^{-b \cdot e^{-cx}} \text{ where } a, b, \text{ and } c \text{ are constants.}$$

- Find the relative rate of change formula for the generic Gompertz function.
- Use a. to find the relative rate of change of a population in $x = 20$ months when $a = 204$, $b = 0.0198$, and $c = 0.15$.
- Briefly interpret what the result of b. means.

For the following exercises, use the population of New York City from 1790 to 1860, given in the following table.

Years since 1790	Population
0	33,131
10	60,515
20	96,373
30	123,706
40	202,300
50	312,710
60	515,547
70	813,669

Table 3.8 New York City

Population Over Time Source:

http://en.wikipedia.org/wiki/Largest_cities_in_the_United_States

_by_population_by_decade.

- 363. [T]** Using a computer program or a calculator, fit a growth curve to the data of the form $p = ab^t$.

- 364. [T]** Using the exponential best fit for the data, write a table containing the derivatives evaluated at each year.

- 365. [T]** Using the exponential best fit for the data, write a table containing the second derivatives evaluated at each year.

- 366. [T]** Using the tables of first and second derivatives and the best fit, answer the following questions:
- Will the model be accurate in predicting the future population of New York City? Why or why not?
 - Estimate the population in 2010. Was the prediction correct from a.?

Chapter Review

Key Terms

acceleration is the rate of change of the velocity, that is, the derivative of velocity

amount of change the amount of a function $f(x)$ over an interval $[x, x + h]$ is $f(x + h) - f(x)$

average rate of change is a function $f(x)$ over an interval $[x, x + h]$ is $\frac{f(x+h)-f(x)}{h}$

chain rule the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function

constant multiple rule the derivative of a constant c multiplied by a function f is the same as the constant multiplied by the derivative: $\frac{d}{dx}(cf(x)) = cf'(x)$

constant rule the derivative of a constant function is zero: $\frac{d}{dx}(c) = 0$, where c is a constant

derivative the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative

derivative function gives the derivative of a function at each point in the domain of the original function for which the derivative is defined

difference quotient of a function $f(x)$ at a is given by

$$\frac{f(a+h)-f(a)}{h} \text{ or } \frac{f(x)-f(a)}{x-a}$$

difference rule the derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g : $\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$

differentiable at a a function for which $f'(a)$ exists is differentiable at a

differentiable function a function for which $f'(x)$ exists is a differentiable function

differentiable on S a function for which $f'(x)$ exists for each x in the open set S is differentiable on S

differentiation the process of taking a derivative

higher-order derivative a derivative of a derivative, from the second derivative to the n th derivative, is called a higher-order derivative

implicit differentiation is a technique for computing $\frac{dy}{dx}$ for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for $\frac{dy}{dx}$

instantaneous rate of change the rate of change of a function at any point along the function a , also called $f'(a)$, or the derivative of the function at a

logarithmic differentiation is a technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly

marginal cost is the derivative of the cost function, or the approximate cost of producing one more item

marginal profit is the derivative of the profit function, or the approximate profit obtained by producing and selling one more item

marginal revenue is the derivative of the revenue function, or the approximate revenue obtained by selling one more item

population growth rate is the derivative of the population with respect to time

power rule the derivative of a power function is a function in which the power on x becomes the coefficient of the term and the power on x in the derivative decreases by 1: If n is an integer, then $\frac{d}{dx}x^n = nx^{n-1}$

product rule the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function: $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + g'(x)f(x)$

quotient rule the derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function: $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$

speed is the absolute value of velocity, that is, $|v(t)|$ is the speed of an object at time t whose velocity is given by $v(t)$

sum rule the derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g : $\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$

Key Equations

Difference quotient	$Q = \frac{f(x)-f(a)}{x-a}$
Difference quotient with increment h	$Q = \frac{f(a+h)-f(a)}{a+h-a} = \frac{f(a+h)-f(a)}{h}$
Slope of tangent line	$m_{\tan} = \lim_{x \rightarrow a} \frac{f(x)-f(a)}{x-a}$ $m_{\tan} = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$
Derivative of $f(x)$ at a	$f'(a) = \lim_{x \rightarrow a} \frac{f(x)-f(a)}{x-a}$ $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$
Average velocity	$v_{\text{ave}} = \frac{s(t)-s(a)}{t-a}$
Instantaneous velocity	$v(a) = s'(a) = \lim_{t \rightarrow a} \frac{s(t)-s(a)}{t-a}$
The derivative function	$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$
Derivative of sine function	$\frac{d}{dx}(\sin x) = \cos x$
Derivative of cosine function	$\frac{d}{dx}(\cos x) = -\sin x$
Derivative of tangent function	$\frac{d}{dx}(\tan x) = \sec^2 x$
Derivative of cotangent function	$\frac{d}{dx}(\cot x) = -\csc^2 x$
Derivative of secant function	$\frac{d}{dx}(\sec x) = \sec x \tan x$
Derivative of cosecant function	$\frac{d}{dx}(\csc x) = -\csc x \cot x$
The chain rule	$h'(x) = f'(g(x))g'(x)$
The power rule for functions	$h'(x) = n(g(x))^{n-1}g'(x)$
Inverse function theorem	$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ whenever $f'(f^{-1}(x)) \neq 0$ and $f(x)$ is differentiable.
Power rule with rational exponents	$\frac{d}{dx}(x^{m/n}) = \frac{m}{n}x^{(m/n)-1}$.
Derivative of inverse sine function	$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$
Derivative of inverse cosine function	$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$

Derivative of inverse tangent function	$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+(x)^2}$
Derivative of inverse cotangent function	$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+(x)^2}$
Derivative of inverse secant function	$\frac{d}{dx} \sec^{-1} x = \frac{1}{ x \sqrt{(x)^2-1}}$
Derivative of inverse cosecant function	$\frac{d}{dx} \csc^{-1} x = \frac{-1}{ x \sqrt{(x)^2-1}}$
Derivative of the natural exponential function	$\frac{d}{dx} (e^{g(x)}) = e^{g(x)} g'(x)$
Derivative of the natural logarithmic function	$\frac{d}{dx} (\ln g(x)) = \frac{1}{g(x)} g'(x)$
Derivative of the general exponential function	$\frac{d}{dx} (b^{g(x)}) = b^{g(x)} g'(x) \ln b$
Derivative of the general logarithmic function	$\frac{d}{dx} (\log_b g(x)) = \frac{g'(x)}{g(x) \ln b}$

Key Concepts

3.1 Defining the Derivative

- The slope of the tangent line to a curve measures the instantaneous rate of change of a curve. We can calculate it by finding the limit of the difference quotient or the difference quotient with increment h .
- The derivative of a function $f(x)$ at a value a is found using either of the definitions for the slope of the tangent line.
- Velocity is the rate of change of position. As such, the velocity $v(t)$ at time t is the derivative of the position $s(t)$ at time t . Average velocity is given by

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}.$$

Instantaneous velocity is given by

$$v(a) = s'(a) = \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a}.$$

- We may estimate a derivative by using a table of values.

3.2 The Derivative as a Function

- The derivative of a function $f(x)$ is the function whose value at x is $f'(x)$.
- The graph of a derivative of a function $f(x)$ is related to the graph of $f(x)$. Where $f(x)$ has a tangent line with positive slope, $f'(x) > 0$. Where $f(x)$ has a tangent line with negative slope, $f'(x) < 0$. Where $f(x)$ has a horizontal tangent line, $f'(x) = 0$.
- If a function is differentiable at a point, then it is continuous at that point. A function is not differentiable at a point if it is not continuous at the point, if it has a vertical tangent line at the point, or if the graph has a sharp corner or cusp.
- Higher-order derivatives are derivatives of derivatives, from the second derivative to the n th derivative.

3.3 Differentiation Rules

- The derivative of a constant function is zero.
- The derivative of a power function is a function in which the power on x becomes the coefficient of the term and the power on x in the derivative decreases by 1.
- The derivative of a constant c multiplied by a function f is the same as the constant multiplied by the derivative.
- The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the

derivative of g .

- The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g .
- The derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.
- The derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function.
- We used the limit definition of the derivative to develop formulas that allow us to find derivatives without resorting to the definition of the derivative. These formulas can be used singly or in combination with each other.

3.4 Derivatives as Rates of Change

- Using $f(a+h) \approx f(a) + f'(a)h$, it is possible to estimate $f(a+h)$ given $f'(a)$ and $f(a)$.
- The rate of change of position is velocity, and the rate of change of velocity is acceleration. Speed is the absolute value, or magnitude, of velocity.
- The population growth rate and the present population can be used to predict the size of a future population.
- Marginal cost, marginal revenue, and marginal profit functions can be used to predict, respectively, the cost of producing one more item, the revenue obtained by selling one more item, and the profit obtained by producing and selling one more item.

3.5 Derivatives of Trigonometric Functions

- We can find the derivatives of $\sin x$ and $\cos x$ by using the definition of derivative and the limit formulas found earlier. The results are

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} \cos x = -\sin x.$$

- With these two formulas, we can determine the derivatives of all six basic trigonometric functions.

3.6 The Chain Rule

- The chain rule allows us to differentiate compositions of two or more functions. It states that for $h(x) = f(g(x))$,

$$h'(x) = f'(g(x))g'(x).$$

In Leibniz's notation this rule takes the form

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

- We can use the chain rule with other rules that we have learned, and we can derive formulas for some of them.
- The chain rule combines with the power rule to form a new rule:

$$\text{If } h(x) = (g(x))^n, \text{ then } h'(x) = n(g(x))^{n-1}g'(x).$$

- When applied to the composition of three functions, the chain rule can be expressed as follows: If $h(x) = f(g(k(x)))$, then $h'(x) = f'(g(k(x)))g'(k(x))k'(x)$.

3.7 Derivatives of Inverse Functions

- The inverse function theorem allows us to compute derivatives of inverse functions without using the limit definition of the derivative.
- We can use the inverse function theorem to develop differentiation formulas for the inverse trigonometric functions.

3.8 Implicit Differentiation

- We use implicit differentiation to find derivatives of implicitly defined functions (functions defined by equations).
- By using implicit differentiation, we can find the equation of a tangent line to the graph of a curve.

3.9 Derivatives of Exponential and Logarithmic Functions

- On the basis of the assumption that the exponential function $y = b^x$, $b > 0$ is continuous everywhere and differentiable at 0, this function is differentiable everywhere and there is a formula for its derivative.
- We can use a formula to find the derivative of $y = \ln x$, and the relationship $\log_b x = \frac{\ln x}{\ln b}$ allows us to extend our differentiation formulas to include logarithms with arbitrary bases.
- Logarithmic differentiation allows us to differentiate functions of the form $y = g(x)^{f(x)}$ or very complex functions by taking the natural logarithm of both sides and exploiting the properties of logarithms before differentiating.

Review Exercises

True or False? Justify the answer with a proof or a counterexample.

367. Every function has a derivative. 368. A continuous function has a continuous derivative. 369. A continuous function has a derivative.
370. If a function is differentiable, it is continuous.

Use the limit definition of the derivative to exactly evaluate the derivative.

371. $f(x) = \sqrt{x+4}$ 372. $f(x) = \frac{3}{x}$

Find the derivatives of the following functions.

373. $f(x) = 3x^3 - \frac{4}{x^2}$ 374. $f(x) = (4 - x^2)^3$ 375. $f(x) = e^{\sin x}$
376. $f(x) = \ln(x+2)$ 377. $f(x) = x^2 \cos x + x \tan(x)$ 378. $f(x) = \sqrt{3x^2 + 2}$
379. $f(x) = \frac{x}{4} \sin^{-1}(x)$ 380. $x^2 y = (y+2) + xy \sin(x)$

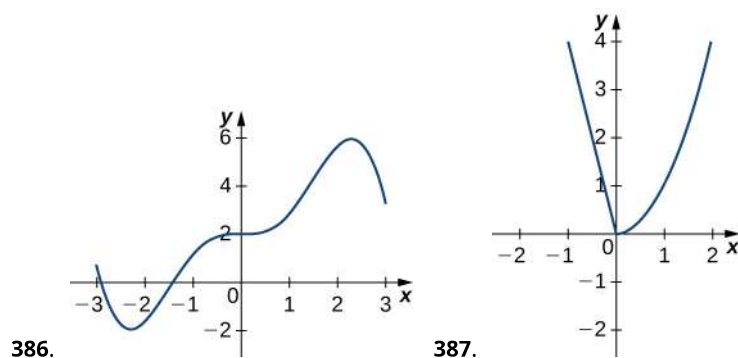
Find the following derivatives of various orders.

381. First derivative of $y = x \ln(x) \cos x$ 382. Third derivative of $y = (3x+2)^2$ 383. Second derivative of $y = 4^x + x^2 \sin(x)$

Find an equation of the tangent line to the following equations at the specified point.

384. $y = \cos^{-1}(x) + x$ at $x = 0$ 385. $y = x + e^x - \frac{1}{x}$ at $x = 1$

Draw the derivative for the following graphs.



The following questions concern the water level in Ocean City, New Jersey, in January, which can be approximated by $w(t) = 1.9 + 2.9 \cos\left(\frac{\pi}{6}t\right)$, where t is measured in hours after midnight, and the height is measured in feet.

388. Find and graph the derivative. What is the physical meaning?
389. Find $w'(3)$. What is the physical meaning of this value?

The following questions consider the wind speeds of Hurricane Katrina, which affected New Orleans, Louisiana, in August 2005. The data are displayed in a table.

Hours after Midnight, August 26	Wind Speed (mph)
1	45
5	75
11	100
29	115
49	145
58	175
73	155
81	125
85	95
107	35

Table 3.9 Wind Speeds of Hurricane Katrina Source:
http://news.nationalgeographic.com/news/2005/09/0914_050914_katrina_timeline.html.

- 390.** Using the table, estimate the derivative of the wind speed at hour 39. What is the physical meaning?
- 391.** Estimate the derivative of the wind speed at hour 83. What is the physical meaning?